Master's Thesis in
**Quantum Science & Technology**

# Distribution of Atom-Photon Entanglement in a Metropolitan Quantum Link


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# Abstract

In recent years, quantum networks have become a major focus of research due to their potential to address scalability issues of current quantum computers and enable provably secure communication between multiple parties.

In this thesis, initial steps towards a quantum link, which is a fundamental building block of a quantum network, are presented. The quantum link consists of two $^{87}\mathrm{Rb}$-atoms, one at Ludwig Maximilian University (LMU) in the center of Munich and one at the Max Planck Institute of Quantum Optics (MPQ) in Garching. Both locations are connected by two optical telecom fibers that span $23.7$ km. First, the fibers were investigated using an optical time-domain reflectometer measurement, which did not reveal any faults in the fibers. The attenuation of each fiber was determined to be below $9.15$ dB, where the fiber with lower loss was chosen as the quantum channel. Additionally, polarization dependent losses below $0.11$ dB were measured using the Müller-matrix method, which is motivated theoretically. Next, the polarization drift was determined by measuring the output polarization of classical light through each fiber over $48$ h. From this measurement, the resulting fidelity drift of a polarization qubit was inferred. Moreover, defining a criterion that a fiber is passively stable as long as the fidelity of the qubit stays above $F \geq 0.99$, a stability of approximately an hour was found for both fibers. In another experiment, photons that were entangled with an atom at LMU were analysed in two different bases at MPQ, while scanning the atomic basis at LMU. To run this experiment, a filter cavity setup was built to filter noise coming from the quantum frequency converters that converted photons between infrared and telecom wavelengths to allow for efficient transmission through the telecom fibers. The external efficiency for the filter cavity setup was determined to be $T_{ext} = (75.14 \pm 1.17)\,\%$, which was found to be mainly limited by fiber coupling. Finally, the fidelity of the experimental state to a maximally entangled Bell state was measured in two different bases and was found to be $\bar{F} = 0.814 \pm 0.031$, which verified entanglement as it was above the entanglement criterion of $F > 0.5$ in both bases. Additionally, the entanglement is measured at a rate of $r = 1.014 \pm 0.053\,\mathrm{min}^{-1}$. Considering the error bars, these are in good agreement with theoretical predictions of $F_{\mathrm{th}} = 0.807$ and $r_{\mathrm{th}} = 1.04\,\mathrm{min}^{-1}$, where the model for fidelity is based on the assumption that the photonic state becomes totally uncorrelated due to noisy detection.

To move towards the goal of atom-atom entanglement, the next step involves the storage of the entangled photon in the Garching node and the characterization thereof. Additionally, experiments in quantum key distribution and quantum teleportation are proposed as an outlook to evaluate practical applications of the quantum link, while quantum steering could be investigated for foundational research in physics.

# Contents

# Chapter 1

# Introduction

The groundbreaking results from foundational quantum physics research have led to the development of various new technologies. One prominent example is magnetic resonance imaging (MRI), a non-invasive imaging technique based on nuclear spins. The spin property of a particle was first observed and described by Stern and Gerlach in 1922 [1].
Another example is Light Amplification by Stimulated Emission of Radiation (Laser), which is for example used in medicine for eye surgery and cancer treatment or in retail for scanning bar codes. The groundwork for this technology was laid out by Einstein in his paper about stimulated emission in 1916 [2].
These technologies, where we as humans played the role of passive observers of quantum mechanical effects, are called "first-generation quantum technologies" or "quantum 1.0" [3].

Nowadays researchers are investigating "second-generation quantum technologies" or "quantum 2.0", where we as humans now actively manipulate and create quantum states to build useful technologies [3]. The European Union identifies four categories of second-generation quantum technologies within their "Quantum Manifesto": Quantum Communication, Quantum Sensing, Quantum Simulation and Quantum Computing [4], which hold significant potential for future applications beyond what is classically possible [5, 6, 7].

A quantum network that Reiserer and Rempe define as "a coherent few- or many-body systems with genuine quantum properties such as entanglement" [8] is another second generation quantum technology. Quantum networks could enable many applications such as measurement precision below the standard quantum limit [9], overcoming scalability issues of current quantum computers [10] and provably secure communication between multiple parties [11].
Strictly speaking, a quantum network consists of at least three so-called nodes [12], which are connected by quantum channels that allow for exchange of quantum information between the nodes. A system with only two nodes and one quantum channel is called a quantum link. In the most general case, a node is a quantum computer [12]. In the simplest case, a node is a device capable of preparing and transmitting arbitrary quantum states as well as performing any one-qubit projective measurement [12].

There are currently many different hardware platforms under investigation for realizing a quantum network node [7]. In this work, trapped neutral $^{87}\mathrm{Rb}$ atoms are used as the stationary nodes as they are identical by nature, making them highly scalable compared to other platforms [13].
Additionally, atoms have a natural and efficient interface to photons when placing them inside of cavities [14]. This interface is important, as photons are the most suitable quantum information carrier [15] over long distances using existing telecom fiber technology to enable interaction between different nodes.
In this thesis, the initial steps towards building a quantum link made out of two $^{87}\mathrm{Rb}$ atoms, one at Ludwig Maximilian University (LMU) in the center of Munich and one at Max Planck Institute of Quantum Optics (MPQ) in Garching, are presented. This includes the characterization of the two optical fibers acting as the quantum and classical channel as well as the verification of atom-photon entanglement between the two nodes of the link.

The thesis is structured as follows: In chapter 2, the reader will be introduced to the concept of entanglement and the theory behind the measurement used to verify entanglement in the experiment.
Chapter 3 introduces the optical fiber network and its characteristics important for distributing photons entangled in their polarization degree of freedom. Specifically, these measurements include the loss through the fiber, the polarization dependent loss and the polarization drift in time. The chapter will be closed by discussing the results in comparison to other existing optical fiber links for quantum links and networks.
Chapter 4 discusses the overall setup idea of the final quantum link, as well as the characterisations thereof, such as the filter cavity setup and detection setup. Moreover, the main result of this thesis, namely atom-photon entanglement over the link will be presented and discussed. A theoretical model predicting the entangling rate as well as the fidelity will also be presented and compared to experimental results. Once again, the results are compared to other existing quantum links.
In the final chapter 5, the results of the aforementioned chapters will be summarized and an outlook on follow-up experiments to explore the quantum link's capabilities will be given.

# Chapter 2

# Theoretical Background

Entanglement is a fundamental resource for enabling applications of quantum networks such as distributed quantum sensing [16] and communication [11]. Therefore, this chapter aims to establish the theoretical tools necessary to analyze and quantify entanglement.

Entanglement was discussed as early as the 1930s, where Einstein, Podolsky and Rosen, published an infamous paper arguing that the "description of reality as given by a wavefunction is not complete" [17].
To construct a complete theory of quantum mechanics as imagined by Einstein, Podolsky and Rosen, one introduces a so-called local hidden variable theory. If a physical system fulfills locality, any action at distance $l$ can occur earliest after the time light needs to travel this distance $t = l/c$. Hidden variables, are variables that cannot be measured in the experiment, therefore hidden, but determine the outcome of the experiment.
To test these theories, Bell proposed an inequality in 1964 [18] that is fulfilled for any local hidden variable theory. Later on, experiments using entangled states showed violation of this inequality, proving that quantum mechanics cannot be described by local hidden variable theories. The main contributors to these experiments at the time Aspect, Clauser and Zeilinger, were later honored “for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science” with the Nobel prize in physics [19], which underlined the novelty and importance of entanglement.

In the following, entanglement will be defined in section 2.1. In section 2.2 local measurements as well as the most general measurement of a single qubit will be introduced. Afterwards, these two concepts are put together to derive the correlator and its visibility of a two qubit state. Then, a theoretical model of the experimental state and its fidelity with the ideally prepared Bell state will be defined in section 2.3. Finally, in section 2.4, all of the previous sections are used to derive a criterion for determining whether a state is entangled based on its fidelity with the ideally prepared Bell state. This criterion is then later applied within chapter 4 of this thesis.

## 2.1 Entanglement Definition

A bipartite state $\rho$ is called entangled if and only if

$$\rho \neq \sum_i p_i \rho_i^A \otimes \rho_i^B, \tag{2.1}$$

where $p_i$ is the probability of measuring the state in $\rho_i^A \otimes \rho_i^B$ with $\sum_i p_i = 1$. Moreover, $\rho_i^A$ is the i-th density matrix acting on the Hilbert space $\mathcal{H}_A$ of subsystem A and $\rho_i^B$ is the i-th density matrix acting on the Hilbert space $\mathcal{H}_B$ of subsystem B. On the other hand, if instead some density matrix is equal to the right hand side of equation 2.1 as follows

$$\rho = \sum_i p_i \rho_i^A \otimes \rho_i^B, \tag{2.2}$$

the state is called separable [20].
In principle, it can be checked whether an arbitrary density matrix is separable by trying to find a transformation, such that it can be written according to equation 2.2. However, doing so is a computationally hard problem. Luckily, using the Peres-Horodecki criterion introduced in section 2.4, it becomes straightforward to proof whether an arbitrary mixed state is entangled or not.

## 2.2 From Correlators to Visibility

To verify entanglement experimentally, local operators on both subsystems will be measured, which are here assumed to be qubits. A local operator on two systems A and B making up a composite system, is given by the following [21]

$$O = O^A \otimes O^B, \tag{2.3}$$

where the superscript A refers to an operator acting only on subsystem A and the superscript B to an operator acting only on subsystem B respectively.
For now, assume that system A is the atom and system B is the photonic state emitted from the atom. Let $\mathcal{B}_A$ and $\mathcal{B}_B$ denote the basis on the respective subsystems, which are:

$$\mathcal{B}_A = \{\, |\uparrow\rangle, |\downarrow\rangle \,\} = \{\, |F = 1, m_f = 1\rangle, |F = 1, m_f = -1\rangle \,\}, \tag{2.4}$$
$$\mathcal{B}_B = \{\, |L\rangle, |R\rangle \,\} = \{\, |S = 1, m_s = 1\rangle, |S = 1, m_s = -1\rangle \,\} \tag{2.5}$$

Here $F$ is the hyperfine angular momentum consisting of the orbital angular momentum and spin of the electron as well as the nuclear spin. $m_f$ is the corresponding hyperfine magnetic quantum number of the atom. Then $S$ is the photon's spin and the corresponding magnetic quantum number is $m_s$.
These are the bases of the experiment, since ideally the excitation shown in figure 4.1 and collection scheme only allow the photon to be in the final states given by $\mathcal{B}_B$, as discussed

at the start of section 4.1. Therefore, the corresponding atomic states are fully described by $\mathcal{B}_A$, since $\mathcal{B}_A$ contains all possible angular momentum states the atom can be found in when a photon is collected into the fiber. Moreover both systems are qubits, since their state space is two-dimensional dim$(\mathcal{B}_A) =$ dim$(\mathcal{B}_B) = 2$.
Coming back to the description of the measurement, the most general measurement on a qubit, can be described by using the Pauli matrices as a basis for the operator space, such that [22]:

$$\sigma_{(\theta,\phi)} = \sin\theta\cos\phi\sigma_x + \sin\theta\sin\phi\sigma_y + \cos\theta\sigma_z, \tag{2.6}$$

where $\sigma_i$ is the i-th Pauli matrix and $(\theta, \phi)$ are parameters characterizing the measurement basis. For example if $(\theta, \phi) = (0, 0)$, then $\sigma_{(\theta=0,\phi=0)} = \sigma_z$, which is a measurement in the Z-basis.
These operators are used to describe the measurement in chapter 4, where the measurement basis for one subsystem is rotated in between measurements. Moreover, these general operators can be expressed further with projection operators as

$$\sigma_{(\theta,\phi)} = P^+_{\theta,\phi} - P^-_{\theta,\phi}, \tag{2.7}$$

where $P^+_{\theta,\phi}$ is the projector onto the eigenstate $\left|\psi^+_{\theta,\phi}\right\rangle = \cos\frac{\theta}{2}\left|\uparrow\right\rangle + e^{i\phi}\sin\frac{\theta}{2}\left|\downarrow\right\rangle$ and $P^-_{\theta,\phi}$ is the projector onto the eigenstate $\left|\psi^-_{\theta,\phi}\right\rangle = \sin\frac{\theta}{2}\left|\uparrow\right\rangle - e^{i\phi}\cos\frac{\theta}{2}\left|\downarrow\right\rangle$ of $\sigma_{(\theta,\phi)}$ respectively. Therefore, the projectors can also be expressed as:

$$P^+_{\theta,\phi} = \left|\psi^+_{\theta,\phi}\right\rangle\left\langle\psi^+_{\theta,\phi}\right|$$
$$P^-_{\theta,\phi} = \left|\psi^+_{\theta,\phi}\right\rangle\left\langle\psi^-_{\theta,\phi}\right|$$

Here the projection operators are described with the example for a state given in the atomic basis from equation 2.4.
If the projection is applied to the photonic part of the composite state, $\left|\uparrow\right\rangle$ and $\left|\downarrow\right\rangle$ have to be replaced with $\left|L\right\rangle$ and $\left|R\right\rangle$ respectively.

Calculating the expectation value of general measurements from equation 2.6 that act locally as given by equation 2.3 on the composite $\rho$, the following expression is obtained

$$K(\theta, \phi, \theta', \phi') = \mathrm{Tr}\big\{\rho(\sigma_{(\theta,\phi)} \otimes \sigma_{(\theta',\phi')})\big\}, \tag{2.8}$$

where $K(\theta, \phi, \theta', \phi')$ is called the correlation [23] or correlator. The angles $(\theta, \phi)$ characterize the measurement on subsystem A while the angles $(\theta', \phi')$ characterize the measurement on subsystem B respectively.
Using equation 2.7, the expression simplifies as follows

$$K(\theta, \phi, \theta', \phi') = \mathrm{Tr}\Big\{\rho(P^+_{\theta,\phi} \otimes P^+_{\theta',\phi'} - P^+_{\theta,\phi} \otimes P^-_{\theta',\phi'} - P^-_{\theta,\phi} \otimes P^+_{\theta',\phi'} + P^-_{\theta,\phi} \otimes P^-_{\theta',\phi'})\Big\}. \tag{2.9}$$

By definition $\mathrm{Tr}\Big\{\rho(P^i_{\theta,\phi} \otimes P^j_{\theta',\phi'})\Big\} = p^{(ij)}_{\theta,\phi,\theta',\phi'}$, where $p^{(ij)}_{\theta,\phi,\theta',\phi'}$ is the probability to project the state onto the composite state $\left|\psi^i_{\theta,\phi}\right\rangle \otimes \left|\psi^j_{\theta',\phi'}\right\rangle$ when measuring $\sigma_{(\theta,\phi)} \otimes \sigma_{(\theta',\phi')}$. Neglecting indices for simplicity of notation, the correlator becomes

$$K(\theta, \phi, \theta', \phi') = p^{(++)} - p^{(+-)} - p^{(-+)} + p^{(--)}. \tag{2.10}$$

Since $p^{(ij)}$ is a probability, the correlator is bounded as follows $K(\theta, \phi, \theta', \phi') \in [-1, 1]$. A value of $-1$ indicates perfect anti-correlation, while a value of $1$ indicates perfect correlation and $0$ indicates no correlation at all between the two subsystems.
In the following, the correlator is shifted such that the possible values become $K'(\theta, \phi, \theta', \phi') \in [0, 1]$, where $K'(\theta, \phi, \theta', \phi')$ is the shifted correlator. This is done such that the visibility, which is defined later in equation 2.13, is not divergent anymore for $\max_\theta(K'(\theta, \phi, \theta', \phi')) = 1$ and $\min_\theta(K'(\theta, \phi, \theta', \phi')) = -1$. The shifted correlator is defined as follows

$$\begin{aligned}
K'(\theta, \phi, \theta', \phi') &= \frac{K(\theta, \phi, \theta', \phi') + 1}{2} && (2.11)\\
&= \frac{p^{(++)} - p^{(+-)} - p^{(-+)} + p^{(--)} + 1}{2} \\
&= \frac{2p^{(++)} + 2p^{(--)}}{2} \\
&= p^{(++)} + p^{(--)}, && (2.12)
\end{aligned}$$

where in the second step the relation $p^{(++)} + p^{(+-)} + p^{(-+)} + p^{(--)} = 1$ was used.

For any variables of the correlator, a visibility can be defined, which tells us the literal visibility of the maximum to the minimum of a function along the chosen variable. For example, the visibility with regards to the angle $\theta$ is given by

$$V = \frac{\max_\theta(K'(\theta, \phi, \theta', \phi')) - \min_\theta(K'(\theta, \phi, \theta', \phi')}{\max_\theta(K'(\theta, \phi, \theta', \phi')) + \min_\theta(K'(\theta, \phi, \theta', \phi')}, \tag{2.13}$$

where $V$ is bounded between $0$ and $1$, since $K'(\theta, \phi, \theta', \phi')$ is bounded between $0$ and $1$ when excluding the case where $\max_\theta(K'(\theta, \phi, \theta', \phi')) = 0 = \min_\theta(K'(\theta, \phi, \theta', \phi'))$. A visibility of $0$ means, that the function is a constant $K' = c$ that does not show any difference between maximum and minimum values. Therefore, the maxima and minima of this function have "no visibility". On the contrary, a visibility of 1 is achieved for example for a simple sinusoidal oscillation with $\theta$ such as $f(\theta) = \sin^2(\theta)$. In that case, the maxima and minima can be well differentiated and have "high visibility".

## 2.3 Experimental Atom-Photon State Model

In the following section, a model of the experimental state given the experimental circumstances described in chapter 4 is presented.

The state that is ideally prepared due to the excitation scheme of the atom, which can be seen in figure 4.1, and subsequent collection scheme, is the maximally entangled Bell-state [24]

$$\begin{aligned} \left|\phi^{+}\right\rangle &= \frac{1}{\sqrt{2}}(|F=1, m_f=-1\rangle \otimes |S=1, m_s=1\rangle + |F=1, m_f=+1\rangle \otimes |S=1, m_s=-1\rangle) \\ &= \frac{1}{\sqrt{2}}(|\downarrow, L\rangle + |\uparrow, R\rangle \end{aligned} \tag{2.14}$$

However, due to experimental imperfections, it is assumed that with a probability of $(1-p)$ the prepared state is completely mixed, which is given by the identity operator $\mathbb{1}$. With a probability of $p$ on the other hand, the ideal state from equation 2.14 is created. Putting this together, and normalizing, the experimental state is modelled by

$$\rho = p\left|\phi^{+}\right\rangle\left\langle\phi^{+}\right| + (1-p)\frac{\mathbb{1}}{4}. \tag{2.15}$$

To know how close this imperfect state is to the ideal state, one can use the measure of fidelity, which provides information about the projection of two arbitrary states onto each other. By definition, the fidelity between some statistical mixture $\rho$ and some pure state $|\psi\rangle$ is given by [25]

$$F(\rho, |\psi\rangle) = \langle\psi|\,\rho\,|\psi\rangle\,. \tag{2.16}$$

Then plugging the imperfect experimental state from equation 2.15 and the ideally prepared state from equation 2.14 into equation 2.16, the fidelity between the two states is obtained to be

$$F(\rho, \left|\phi^{+}\right\rangle) = \frac{3}{4}p + \frac{1}{4}. \tag{2.17}$$

## 2.4 Entanglement Criterion

To determine when the state from equation 2.15 is entangled, the Peres-Horodecki criterion, PPT criterion in the following, can be used. The PPT criterion is a necessary and in the case of two qubits sufficient condition to proof that a density matrix $\rho$ is separable [26, 27]. The PPT criterion states that, if and only if an eigenvalue of the partially transposed density matrix is negative, the state is entangled. Mathematically, this can be expressed as

$$\rho = \sum_i p_i \rho_i^A \otimes \rho_i^B \iff \forall \lambda_i \in \sigma(\rho^{T_B}) : \lambda_i \geq 0, \tag{2.18}$$

where $\lambda_i$ are the eigenvalues of $\rho^{T_B}$ and $\sigma(\rho^{T_B})$ is the spectrum of $\rho^{T_B}$, which is the set of all eigenvalues of $\rho^{T_B}$. Here $\rho^{T_B}$ is the partial transposition applied to subsystem B of the composite state $\rho$. The partial transposition on a bipartite system is defined by

$$\begin{aligned}\rho^{T_B} &= \sum_{ijkl} c_{ijkl} \left|i\right\rangle \left\langle j\right| \otimes (\left|k\right\rangle \left\langle l\right|)^T \\ &= \sum_{ijkl} c_{ijkl} \left|i\right\rangle \left\langle j\right| \otimes (\left|l\right\rangle \left\langle k\right|). \end{aligned} \tag{2.19}$$

Using the bases from equations 2.4 and 2.5, the following basis for the composite system is constructed

$$\mathcal{B} = \left\{ \left|\uparrow, L\right\rangle, \left|\uparrow, R\right\rangle, \left|\downarrow, L\right\rangle, \left|\downarrow, R\right\rangle \right\}. \tag{2.20}$$

Applying partial transposition, defined in equation 2.19, to the experimental state $\rho$, given by equation 2.15, the resulting state can be represented using the aforementioned basis from equation 2.20 as

$$\rho^{T_B} = \begin{bmatrix} \frac{1-p}{4} & 0 & 0 & \frac{p}{2} \\ 0 & \frac{1+p}{4} & 0 & 0 \\ 0 & 0 & \frac{1+p}{4} & 0 \\ \frac{p}{2} & 0 & 0 & \frac{1-p}{4} \end{bmatrix}. \tag{2.21}$$

Using common tools to find the eigenvalues of matrices, the following eigenvalues are found

$$\begin{aligned}\lambda_1 &= \frac{1+p}{4} = \lambda_2 = \lambda_3, \\ \lambda_4 &= \frac{1-3p}{4}. \end{aligned} \tag{2.22}$$

Out of these eigenvalues, only eigenvalue $\lambda_4$ can become negative if and only if $p > \frac{1}{3}$, as $p$ is a probability.
Therefore, the experimentally prepared state given by equation 2.15, is entangled, if and only if $p > \frac{1}{3}$ according to the PPT criterion 2.18. This statement can be analogously formulated using the fidelity of the imperfect experimental state to the ideal entangled state from equation 2.14. Plugging $p > \frac{1}{3}$ into equation 2.17, the following criterion for evaluating whether $\rho$ is entangled or not emerges:

$$F(\rho, \left|\phi^+\right\rangle) > \frac{1}{2} \iff \rho \text{ entangled} \tag{2.23}$$

In the previous paragraphs, it was shown how to proof that the experimental state is entangled, given that the fidelity between experimental and ideal state can be extracted. In the following, it will be shown how the fidelity relates to the visibility, which is the quantity that is easily accessible in the experiment.
In the experiment a correlation measurement is performed just like described in section 2.2. In the following, it is assumed without loss of generality that in equation 2.11, $\phi = \phi' = \theta' = 0$

and $\theta$ is varied. Thus the correlator with respect to the experimental state from equation 2.15 yields:

$$\begin{aligned}
K'(\theta) &= p_{++} + p_{--} \\
&= \mathrm{Tr}\Big\{\rho(P^{+}_{\theta,\phi=0} \otimes P^{+}_{\theta'=0,\phi'=0} + P^{-}_{\theta,\phi=0} \otimes P^{-}_{\theta'=0,\phi'=0})\Big\} \\
&= p\,\mathrm{Tr}\big\{\big(\left|\phi^+\right\rangle\left\langle\phi^+\right|\big)\big(P^{+}_{\theta} \otimes P^{+} + P^{-}_{\theta} \otimes P^{-}\big)\big\} + \frac{(1-p)}{4}\,\mathrm{Tr}\big\{P^{+}_{\theta} \otimes P^{+} + P^{-}_{\theta} \otimes P^{-}\big\} \\
&= p\,\mathrm{Tr}\big\{\big(\left|\phi^+\right\rangle\left\langle\phi^+\right|\big)\big(P^{+}_{\theta} \otimes P^{+} + P^{-}_{\theta} \otimes P^{-}\big)\big\} + \frac{(1-p)}{2} \\
&= p\left\langle\phi^+\right|\big(P^{+}_{\theta} \otimes P^{+} + P^{-}_{\theta} \otimes P^{-}\big)\left|\phi^+\right\rangle + \frac{(1-p)}{2} \\
&= p\left\langle\phi^+\right|\Bigg(\Big(\cos^2\frac{\theta}{2}\left|\uparrow\right\rangle\left\langle\uparrow\right| + \sin^2\frac{\theta}{2}\left|\downarrow\right\rangle\left\langle\downarrow\right| + \cos\frac{\theta}{2}\sin\frac{\theta}{2}(\left|\uparrow\right\rangle\left\langle\downarrow\right| + \left|\downarrow\right\rangle\left\langle\uparrow\right|)\Big)\left|L\right\rangle\left\langle L\right| \\
&\quad + \Big(\sin^2\frac{\theta}{2}\left|\uparrow\right\rangle\left\langle\uparrow\right| + \cos^2\frac{\theta}{2}\left|\downarrow\right\rangle\left\langle\downarrow\right| - \cos\frac{\theta}{2}\sin\frac{\theta}{2}(\left|\uparrow\right\rangle\left\langle\downarrow\right| + \left|\downarrow\right\rangle\left\langle\uparrow\right|)\Big)\left|R\right\rangle\left\langle R\right|\Bigg)\left|\phi^+\right\rangle \\
&\quad + \frac{(1-p)}{2} \\
&= p\sin^2\frac{\theta}{2} + \frac{(1-p)}{2}
\end{aligned} \tag{2.24}$$

By plugging this back into the visibility in equation 2.13, the visibility becomes

$$\begin{aligned}
V &= \frac{\max_\theta\left(p\sin^2\frac{\theta}{2} + \frac{(1-p)}{2}\right) - \min_\theta\left(p\sin^2\frac{\theta}{2} + \frac{(1-p)}{2}\right)}{\max_\theta\left(p\sin^2\frac{\theta}{2} + \frac{(1-p)}{2}\right) + \min_\theta\left(p\sin^2\frac{\theta}{2} + \frac{(1-p)}{2}\right)} \\
&= \frac{p + \frac{(1-p)}{2} - \frac{(1-p)}{2}}{p + \frac{(1-p)}{2} + \frac{(1-p)}{2}} \\
&= p
\end{aligned} \tag{2.25}$$

Plugging this relation back into 2.17, the fidelity can be expressed as

$$F = \frac{3}{4}V + \frac{1}{4}. \tag{2.26}$$

To estimate a lower bound for the fidelity of the experimental state, it is assumed that the atom can be in a third state. This accounts for the fact that the storage scheme is imperfect and the atom may populate the $\left|F = 1, m_f = 0\right\rangle$ ground state due to uncompensated magnetic fields along other directions than the quantization axis [28].
For the following $(2 \times 3)$-dimensional composite state, which is constructed with analogous arguments as the $(2 \times 2)$-dimensional state from equation 2.15

$$\rho = p\left|\phi^+\right\rangle\left\langle\phi^+\right| + \frac{1-p}{6}\mathbb{1}, \tag{2.27}$$

the fidelity can be calculated according to equation 2.16 to be $F \geq \frac{5}{6}p + \frac{1}{6}$. Using the relation between visibility $V$ and the probability $p$ from equation 2.25, the lower bound estimate for the fidelity becomes [29]

$$F \geq \frac{5}{6}V + \frac{1}{6} \tag{2.28}$$

Finally, as derived in equation 2.23 under the assumption of the experimental state according to equation 2.15, the state can be claimed to be entangled if and only if the above fidelity exceeds $0.5$.

However, one needs to be careful to claim entanglement, if only one such correlator as shown above is measured, as one specific state was assumed to be present in the experiment. For example, if instead of the assumed experimental state shown in equation 2.15, the following state is present in the experiment

$$|\psi\rangle = |\downarrow, L\rangle\,, \tag{2.29}$$

the correlator with regards to this state becomes in analogy to 2.24

$$\begin{aligned} K'_{\mathrm{cl}}(\theta) &= \langle\downarrow, L|\,(P^{+}_{\theta,\phi=0} \otimes P^{+}_{\theta'=0,\phi'=0} + P^{-}_{\theta,\phi=0} \otimes P^{-}_{\theta'=0,\phi'=0})\,|\downarrow, L\rangle \\ &= \sin^2\frac{\theta}{2}. \end{aligned} \tag{2.30}$$

Thus under the assumption of the experimental state $\rho$ from equation 2.15 even though $|\psi\rangle$ is the state present in the experiment, the visibility according to equation 2.13 would be $V = 1$. The corresponding fidelity according to equation 2.28 would then be $F = 1$, which would be above the criterion for entanglement of $F > 0.5$. As this example illustrates, entanglement can not be verified if only one correlation measurement in one specific basis is used, since classically correlated states such as $|\psi\rangle$ can also show entanglement according to the entanglement criterion.

However, it is known that entangled states show full correlation oscillations in every basis while classically correlated states do not. Therefore, the experimental state can be claimed to be entangled, if two correlation measurements using sufficiently different bases each violate $F > 0.5$ [24].

For example, rotating the atomic basis between $X$, $Y$, $-X$ and $-Y$ in each measurement, while measuring the photon in the H/V-basis in one and in the D/A-basis in another experiment respectively, and finding an average fidelity of $F > 0.5$ is sufficient to claim entanglement according to [24].

# Chapter 3

# Optical Fiber Network

In order to realize a quantum link, one needs two nodes and a connecting quantum channel. In this thesis, the nodes are made of single $^{87}\mathrm{Rb}$-atoms while optical telecom fibers are used as the quantum and classical channel. In the following, the use of optical telecom fibers as quantum channels is motivated.
DiVincenzo's criteria for quantum communication establish requirements which a quantum channel has to fulfill. These are: [30]

6. "The ability to interconvert stationary and flying qubits."
7. "The ability to faithfully transmit flying qubits between specified locations."

Criterion number 7 is the hardest requirement to fulfill for any quantum network, since faithful transmission of flying qubits over long distances is generally not possible due to losses. Unlike to classical information, where a signal can be amplified and thus transmitted despite losses, this is not possible with quantum information due to the no-cloning theorem [31].
Currently, the only viable hardware option for flying qubits are photons [15] as they offer fast and low-loss transmission of quantum information. Due to the limitations of free space transmission in urban environments caused by buildings in the beam path for example, which can only be overcome with costly and time-consuming methods such as satellite communication, transmission through optical fibers is the chosen method in this thesis.
To achieve minimal losses, telecom photons and optical telecom fibers are used. However, interfacing $^{87}\mathrm{Rb}$-atoms with telecom photons is technically much more challenging than infrared photons. Therefore, quantum frequency converters are used to convert emitted photons in the infrared to the telecom band and back. With this interface, requirement number 6 and 7 are fulfilled, making optical telecom fibers a valid choice according to DiVincenzo.

This chapter characterizes the optical telecom fibers properties as quantum channels for transmitting qubits encoded in the polarization of single photons. First, general information about the two telecom fibers will be given in section 3.1. Then the attenuation is presented in section 3.2 and related polarization dependent losses in section 3.3. Next, polarization stability of the fibers is discussed in section 3.4 and finally a short summary and comparison to existing optical fibers used in quantum links and networks is given in section 3.5

## 3.1 Fundamentals

To provide an overall picture of our fiber network, the fundamentals of the two rented fibers from M-net are presented first. One fiber is to be used as the classical and the other is to be used as the quantum channel.

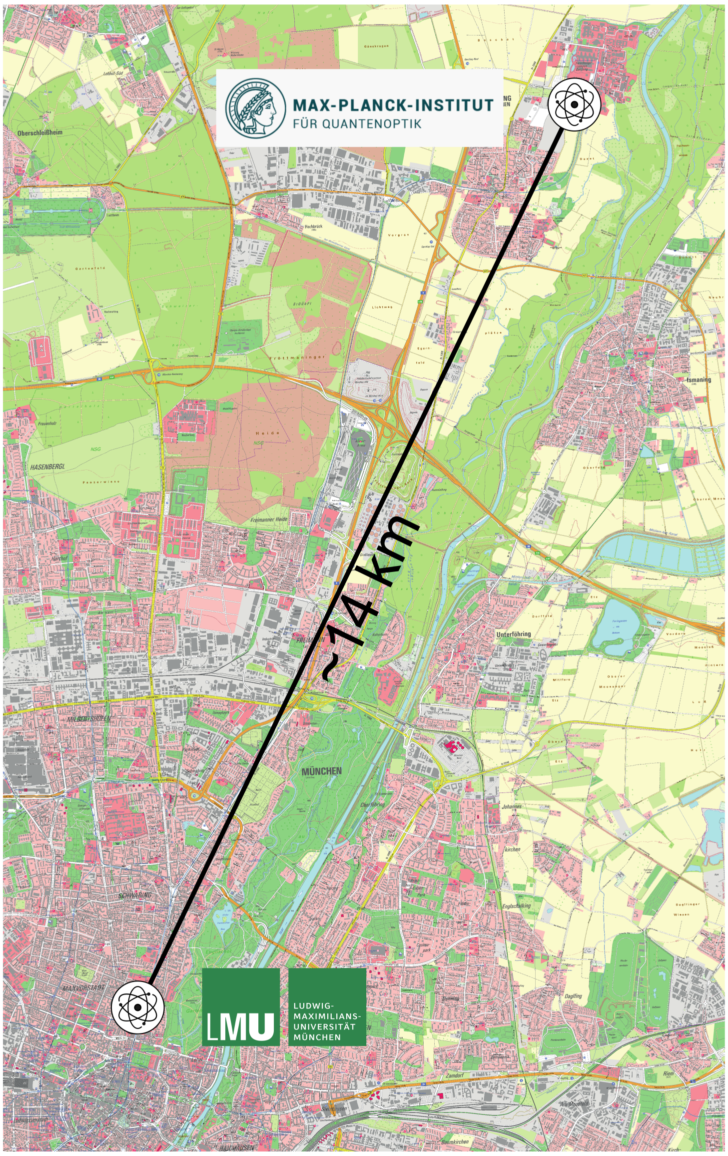



Figure 3.1: Map of the Munich greater area, showing the locations of the quantum link's nodes separated by $14\,\mathrm{km}$. One is located at LMU in the center of Munich, seen in the bottom left, while the other is located at MPQ in Garching-Forschungszentrum, which can be seen in the top right. Map provided by "Bayerische Vermessungsverwaltung - www.geodaten.bayern.de".

The classical channel is used to synchronize the setups, where LMU is given experimental control over the setup at MPQ for atom-photon entanglement in chapter 4 to simplify timing. For example, the detector counts are directly sent to LMU as TTL pulses for analysis. Moreover, the classical channel is multiplexed, allowing different signals such as TTL pulses and RF signals or protocols such as Ethernet to be distributed at different wavelengths through the same glass fiber. To do so, a communication system that is able to convert between optical and electrical signals is used. The quantum channel on the other hand is used for sending quantum information encoded in the polarization of single photons between the nodes - making the link a real quantum link supplemented by a single classical channel.
A layout of the quantum link showing the locations of the two nodes within the greater Munich area can be seen in figure 3.1. The node at LMU is located in the city center of Munich, which is shown in the bottom left of the figure, while the node at MPQ is located in Garching-Forschungszentrum to the top right of the figure. The connecting fibers run mostly along the subway line U6 starting at LMU until Garching. From there they diverge from the subway line and finally connect to MPQ. The main discriminator between these two fibers is the colour of their outer coating that can be seen, when working with them in the laboratory at MPQ. Therefore, one fiber is referred to as the green and the other as the red fiber. The green fiber is used as the quantum and the red fiber as the classical channel as discussed in section 3.2.

To characterize the length and more importantly the quality of the two fibers, an optical time-domain reflectometer (OTDR) is used. An OTDR works by sending light pulses into an optical fiber to record the reflected power over time due to Rayleigh scattering in the fiber. From the time of arrival of the reflected signal, one can locate from where in the fiber the reflected signal came from. In figure 3.2, the reflected power in dB is plotted as a function of the distance in km in the fiber for the red and green fiber with the red and green curve respectively. Using each curve, one can locate splices, connectors and breaks of the fiber, as these show certain features in an OTDR measurement. Moreover, this measurement can be used to determine the length of the fiber, which reveals a length of $23.71$ km for the green and $23.74$ km for the red fiber.
In figure 3.2, some of these features are highlighted using vertical lines. For example the sudden increase in the reflection of the green fiber at roughly $20$ km indicates a connector. At the same position for the red fiber, only a small drop in the reflected power can be observed that continues to stay lower until termination of the fiber, which indicates a splice between a fiber with higher Rayleigh scattering to a fiber with lower Rayleigh scattering. All the other features highlighted in figure 3.2 are also splices and are found at the same position. This is a verification that the fibers are co-propagating since they have either splices or connectors to different fibers at the same intersections.

## 3.2 Attenuation

In the following, the method for measuring the attenuation of the optical fibers as well as the result is presented.

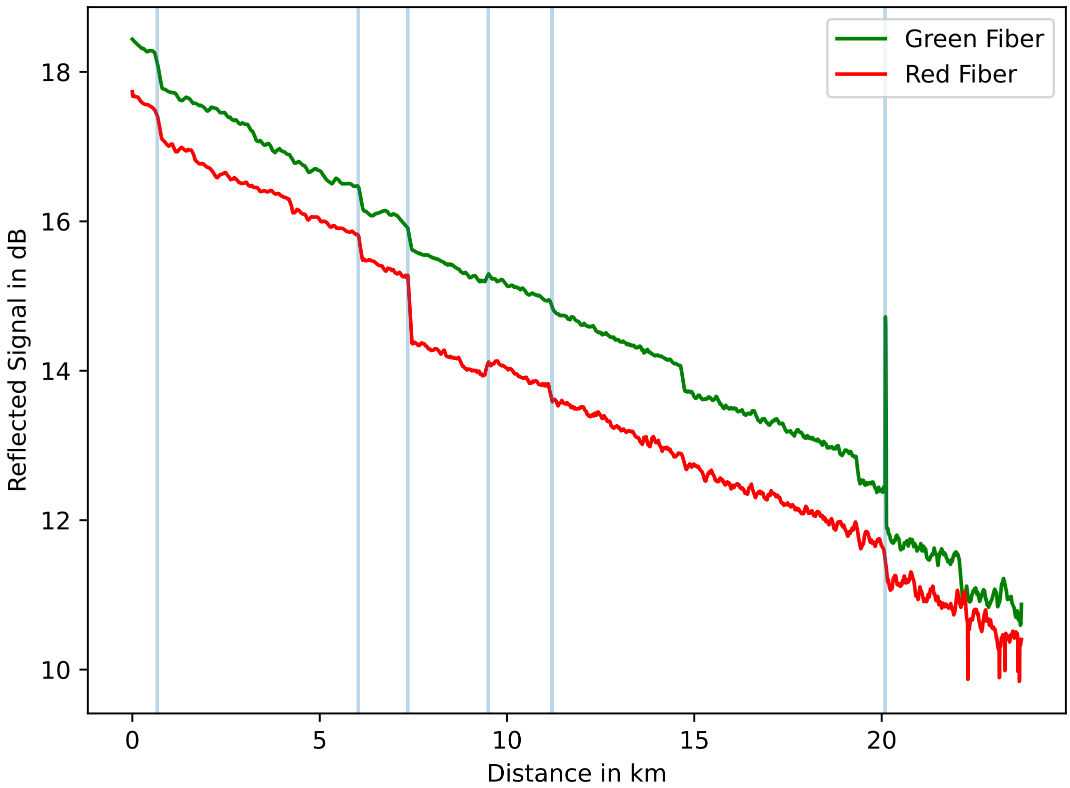



Figure 3.2: The plot shows the reflected signal in dB from an optical time domain reflectometer measurement versus the distance in km from which that signal is reflected. The red curve is measured by sending in a $200$ ns pulse into the red fiber and the green curve represents the same for the green fiber. The vertical lines show points of interest such as splices or connectors, which appear at the same position in the fibers due to their copropagation. Averaging over many OTDR measurements, the green fiber is found to be $23.71$ km and the red fiber is found to be $23.74$ km long.

First, two power meters were calibrated with respect to each other. For details, see the appendix A, where the calibration curves for light at $1514$ nm, at $1550$ nm and at $1610$ nm are shown. Then, LMU prepares light at the wavelength of interest at $1514$ nm, measures the power using one of the calibrated power meters and then sends it through the fiber under test. Finally, the power is also measured at MPQ using the other calibrated power meter.
Using the calibration from figure A.1a, a power of $P_{\mathrm{LMU}} = 3.432$ mW at LMU and a power of $P_{\mathrm{MPQ}} = 503\,\mu$W at MPQ is measured for the green fiber. This yields a transmission of $14.4\,\%$, or equivalently $8.41$ dB.
For the red fiber, one obtains $P_{\mathrm{LMU}} = 3.420$ mW and $P_{\mathrm{MPQ}} = 418\,\mu$W, which implies a transmission of $12.2\,\%$, or equivalently $9.14$ dB.

Also taking into consideration the lengths of the respective fibers, a final attenuation per distance of $0.355\,\frac{\mathrm{dB}}{\mathrm{km}}$ for the green fiber and $0.385\,\frac{\mathrm{dB}}{\mathrm{km}}$ for the red fiber is obtained. These numbers are slightly higher than standard literature values of about $0.3\,\frac{\mathrm{dB}}{\mathrm{km}}$ bare fiber attenuation at $1514$ nm, as the fibers of this thesis have splices and connectors which add losses.

For the quantum channel, the transmission should be as high as possible, as the signal can not be amplified due to the no cloning theorem. Therefore, the green fiber is chosen as the quantum channel. In contrast, the performance of the classical channel is not affected by losses as classical signals can be amplified such that the information is not lost during transmission.

## 3.3 Polarization Dependent Losses

In the previous section, it is shown how much the intensity of light gets lost when sent between LMU and MPQ. However, polarization dependent losses of the fibers are not yet quantified. These impact the fidelity with which quantum information is transmitted, since the qubits are encoded in the polarization of photons. To quantify the polarization dependent loss (PDL), one uses the following definition:

$$\mathrm{PDL} = -10\log\frac{T_{\mathrm{min}}}{T_{\mathrm{max}}} \tag{3.1}$$

Here $T_{\mathrm{min}}$ is the minimum and $T_{\mathrm{max}}$ the maximum transmission of all possible input polarization. To illustrate why this quantity is worth knowing when using the qubit encoding mentioned above, two extreme cases are discussed in the following.
Assume that quantum information is encoded in the $\{|\mathrm{H}\rangle, |\mathrm{V}\rangle\}$ polarization basis of a photon. Moreover, assume that there is extreme polarization dependent losses such that one polarization is completely lost through the fiber. Then according to equation 3.1, this means $T_{\mathrm{min}} = 0$ and the $\mathrm{PDL}$ becomes

$$\mathrm{PDL} \xrightarrow[T_{\mathrm{min}}\to 0]{} \infty.$$

Also assume for the sake of illustration that vertical polarization is the one to be completely lost, while horizontal polarization is transmitted perfectly. Then, the output state of an equal superposition through the fiber can be written as (with renormalization):

$$|\psi_{\mathrm{in}}\rangle = \frac{1}{\sqrt{2}}\Big(|\mathrm{H}\rangle + |\mathrm{V}\rangle\Big) \xrightarrow[\mathrm{Fiber}]{\mathrm{PDL}} |\mathrm{H}\rangle = |\psi_{\mathrm{out}}\rangle \tag{3.2}$$

Therefore the fidelity between input and output state is according to equation 2.16

$$F = \frac{1}{2},$$

which differs from the ideal case, where the fiber shows no polarization dependent loss. This is demonstrated in the next extreme case, where all polarizations are transmitted equally through the fiber: $T_{\min} = T_{\max}$. Therefore by definition from equation 3.5

$$\mathrm{PDL} = 0.$$

In this case of equal transmission of all polarizations, an equal superposition of the photon's polarization is mapped as follows

$$|\psi_{\mathrm{in}}\rangle' = \frac{1}{\sqrt{2}}\Big(|\mathrm{H}\rangle + |\mathrm{V}\rangle\Big) \xrightarrow[\text{Fiber}]{\text{no PDL}} \frac{1}{\sqrt{2}}\Big(|\mathrm{H}\rangle + |\mathrm{V}\rangle\Big) = |\psi_{\mathrm{out}}\rangle' \tag{3.3}$$

Calculating the fidelity of the input and output states, the fidelity is found to be

$$F' = 1,$$

which is considerably different than the fidelity of $F = \frac{1}{2}$ that is obtained when assuming the other extreme case of polarization dependent losses. This comparison demonstrates the effects polarization dependent losses can have on the fidelity of the transmitted polarization qubits.
Therefore, the higher the polarization dependent loss, the lower the fidelity of the transmitted polarization qubits through the fiber.

### 3.3.1 Müller Matrix Method

The Müller matrix method is used to characterize the polarization dependent losses, as the experimental complexity is lower than other methods. For this method, one only has to transmit light with four different polarizations and monitor their input and output power through the fiber under test. The only condition these polarizations have to fulfill is the fact that they need to be linearly independent in the Stokes vector formalism. The white paper that is used as a rough guideline for the following derivations is "Polarization-Dependent Loss $(\mathrm{PDL})$ Measurement Theory" by Viavi Solutions [32].
Essentially, the method works by characterizing the first row of the Müller matrix by transmitting four different polarizations of light through the fiber. Since the first entry of the Stokes vector is the intensity of light, the first row of the Müller matrix describes how the intensity of light changes because of the fiber depending on the input polarization. For example, given an input beam of light $\vec{S}$ being mapped to an output beam of light $\vec{S}'$ by some Müller matrix $\underline{M}$ of the fiber, the mapping is given by

$$\begin{pmatrix} m_{00} & m_{01} & m_{02} & m_{03} \\ m_{10} & m_{11} & m_{12} & m_{13} \\ m_{20} & m_{21} & m_{22} & m_{23} \\ m_{30} & m_{31} & m_{32} & m_{33} \end{pmatrix} \cdot \begin{pmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{pmatrix} = \begin{pmatrix} S_0' \\ S_1' \\ S_2' \\ S_3' \end{pmatrix}$$

By doing the necessary matrix vector calculations for $S_0'$ of the output beam, which describes the intensity of the output beam by definition, the following expression is obtained

$$S_0' = m_{00}S_0 + m_{01}S_1 + m_{02}S_1 + m_{03}S_2. \tag{3.4}$$

As $S_1, S_2$ and $S_3$ completely characterize the polarization of the input light, it becomes apparent how the intensity of the output light is dependent on the polarization of the input light. In accordance to the Müller matrix method, right-circular, diagonal, vertical and horizontal light are chosen as the four input polarizations necessary to characterize polarization dependent loss.
Now according to [32], the polarization dependent loss from equation 3.1 can be equivalently expressed as

$$\mathrm{PDL} = -10\log\left(\frac{m_{00} - \sqrt{m_{01}^2 + m_{02}^2 + m_{03}^2}}{m_{00} + \sqrt{m_{01}^2 + m_{02}^2 + m_{03}^2}}\right), \tag{3.5}$$

where $m_{0\mathrm{i}}$ are entries of the zeroth row and i-th column of the Müller matrix, which is given by

$$\underline{M} = \begin{pmatrix} m_{00} & m_{01} & m_{02} & m_{03} \\ m_{10} & m_{11} & m_{12} & m_{13} \\ m_{20} & m_{21} & m_{22} & m_{23} \\ m_{30} & m_{31} & m_{32} & m_{33} \end{pmatrix}.$$

The first row describes the polarization dependent losses as can be seen from equation 3.4. They can be obtained by measuring the power of the four input polarizations according to [32] as follows:

$$m_{00} = \frac{T_\mathrm{H} + T_\mathrm{V}}{2} \tag{3.6}$$

$$m_{01} = \frac{T_\mathrm{H} - T_\mathrm{V}}{2} \tag{3.7}$$

$$m_{02} = T_\mathrm{D} - \frac{T_\mathrm{H} + T_\mathrm{V}}{2} \tag{3.8}$$

$$m_{03} = \frac{T_\mathrm{H} + T_\mathrm{V}}{2} - T_\mathrm{R} \tag{3.9}$$

Here $T_i$ with $i \in \{H, V, D, R\}$ is the measured transmission for the respective input polarization, which is indicated by the index.

### 3.3.2 Setup

After having discussed how the measurement works in theory in section 3.3.1, the experimental implementation thereof is described in the following.

A schematic picture of how the polarization dependent losses are measured can be found in figure 3.3. To measure them, a fiber-coupled laser is emitted into free space using a fiber coupler. Then, two quarter and one half waveplate are used to set any input polarization.

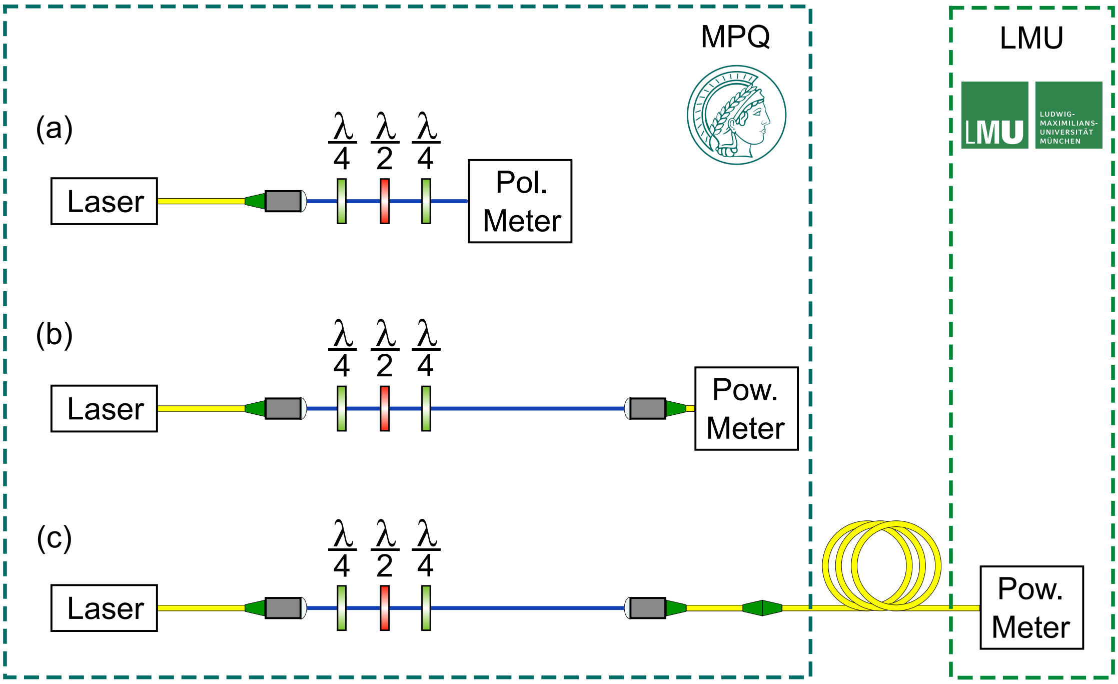



Figure 3.3: The experimental setup used to measure polarization dependent losses of both fibers that connect LMU and MPQ at $\lambda = 1514$ nm. (a) First, the light coming from the laser is emitted into free space to set any polarization using multiple waveplates and monitoring it on a polarimeter (Pol. Meter). (b) The polarimeter is taken out and the light is coupled back into a fiber to measure the incoming power to the fiber with a power meter (Pow. Meter). (c) Then, instead of connecting the fiber to the power meter, it is plugged into the fiber under test leading to LMU, where the power is also measured.

Afterwards, the light is coupled back into a fiber, which is either connected to a power sensor to measure the power of the light or connected to fiber under test. The polarization of the light is monitored by placing a polarimeter in between the last waveplate and the fiber coupler. The experimental sequence is as follows and describes the method depicted in figure 3.3 in more detail:

(a) Place a polarimeter into the beam path at MPQ and use the waveplates to set the desired input polarization.

(b) Take the polarimeter out again and connect the fiber of the outcoupler to the power meter and measure the incoming power over $30$ s at MPQ.

(c) Now connect the aforementioned fiber to one of the two fibers leading to LMU, where the power is again measured for $30$ s.

The time between power measurements at LMU and MPQ is minimized such that no systematic errors like long-term laser power drifts influenced the measurement. Additionally, the error of short scale power fluctuations is given by the variance of the collected data within the $30\,\mathrm{s}$ time frame. The error of the polarization dependent losses is then derived using these errors and Gaussian error propagation.

### 3.3.3 Results and Discussion

The theory as well as the experimental setup for measuring the polarization dependent losses are described in sections 3.3.1 and 3.3.2. In this section, the above sections are used to measure and analyse the polarization dependent loss of the two fibers.

| Polarization | Azimuth | Ellipticity | $P_{\mathrm{LMU}}$ in µW | $P_{\mathrm{MPQ}}$ in µW | T |
|---|---|---|---|---|---|
| R | $78.55°$ | $44.90°$ | $72.35(12)$ | $722.4(1.5)$ | $0.099952(266)$ |
| D | $45.07°$ | $0.11°$ | $71.15(14)$ | $708.2(1.6)$ | $0.100265(300)$ |
| V | $-89.74°$ | $0°$ | $71.41(11)$ | $719.9(1.4)$ | $0.098996(246)$ |
| H | $-0.07°$ | $0.24°$ | $72.99(11)$ | $726.4(1.1)$ | $0.100281(214)$ |

(a) Values for the red fiber yielding a polarization dependent loss of $\mathrm{PDL} = (0.083 \pm 0.024)\,\mathrm{dB}$.

| Polarization | Azimuth | Ellipticity | $P_{\mathrm{LMU}}$ in µW | $P_{\mathrm{MPQ}}$ in µW | T |
|---|---|---|---|---|---|
| R | $82.99°$ | $44.91°$ | $83.08(14)$ | $735.5(1.2)$ | $0.112731(264)$ |
| D | $44.91°$ | $0.05°$ | $84.21(19)$ | $740.7(8)$ | $0.113462(182)$ |
| V | $-89.96°$ | $0.03°$ | $82.57(18)$ | $732.2(1.3)$ | $0.112544(316)$ |
| H | $-0.03°$ | $-0.06°$ | $81.93(11)$ | $712.7(9)$ | $0.114727(211)$ |

(b) Values for the green fiber yielding a polarization dependent loss of $\mathrm{PDL} = (0.109 \pm 0.019)\,\mathrm{dB}$.

Table 3.1: Table showing the power measurements of four different input polarizations for the red and green fiber respectively. The first three columns describe the set input polarization, while columns four and five display the power measured after transmission at LMU and before transmission through the fiber at MPQ respectively. The last column gives the corresponding transmission according to the calibration from the appendix A.

By plugging equations 3.6 to 3.9 into equation 3.5 and using the values from table 3.1, a polarization dependent loss of $\mathrm{PDL} = (0.083 \pm 0.024)\,\mathrm{dB}$ is obtained for the red fiber, while for the green fiber one obtains $\mathrm{PDL} = (0.109 \pm 0.019)\,\mathrm{dB}$. These values are well-aligned with expectations of low polarization dependent losses since the fiber showed great passive stability, see section 3.4, meaning there is negligible influences of heat and mechanical stress from the environment. These results are similar in value to a shorter fiber for which the authors in [33] find $\mathrm{PDL} = (0.08 \pm 0.09)\,\mathrm{dB}$. For their $14\,\mathrm{km}$ long fiber, the authors find a faulty splice and generally lower stability, as parts of it run along the electrical overhead power and is thus subject to wind. This explains why they find a comparable $\mathrm{PDL}$ to the $\mathrm{PDL}$ of the fibers presented here even though their fiber is significantly shorter. Additionally, the polarization dependent losses of their fiber show small fluctuations in time due to the aforementioned influences of the environment. Again, this is not observed for the fibers of this thesis, since they have much greater passive stability.

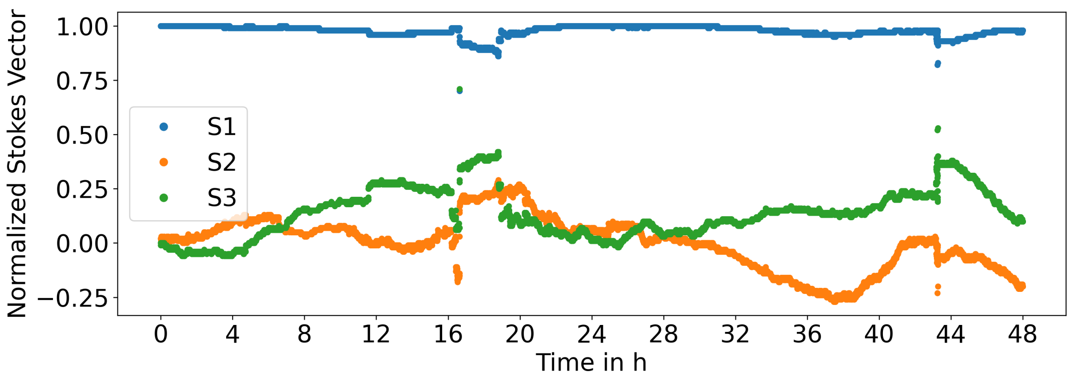
## 3.4 Polarization Drift in Time

Due to changes in the environment of the fibers such as mechanical stress and temperature on different time scales, the birefringence of the fiber starts to drift. If that happens, also the polarization of the transmitted light through the fiber starts to drift. Just as polarization dependent losses, these polarization drifts also have an impact on the fidelity of transmitted qubits, since the quantum information is encoded in the polarization of a single photon. In contrast to polarization dependent losses however, these drifts can be compensated by inducing counteracting mechanical stress on the fiber.

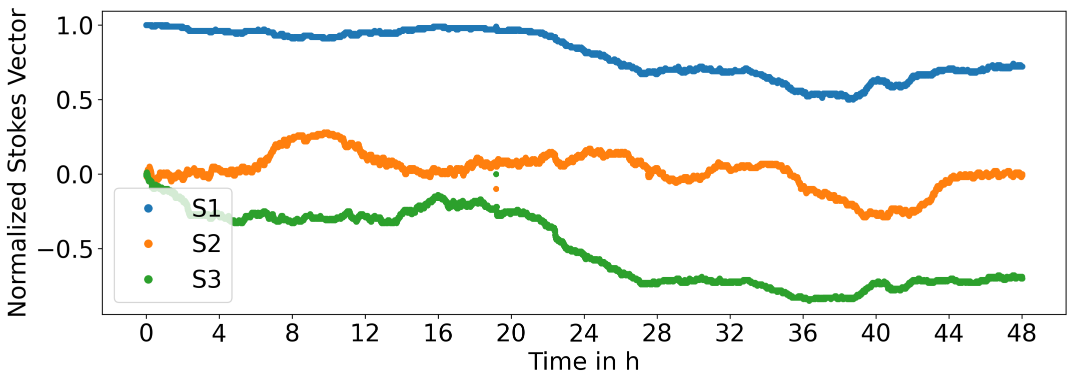
To characterize this drift, the output polarization of classical light is monitored over time for a certain stable input polarization to the fiber under test. The time that the output polarization stays roughly constant (a more detailed criterion is mentioned below) is called passive stability. This passive stability is the figure of interest in this section. The polarization drift can be seen in figure 3.4b for the green and in figure 3.4a for the red fiber. Figure 3.4a, which shows the polarization drift of the red fiber, is discussed in more detail. Its discussion can then be translated to figure 3.4b for the green fiber, since it has an analogous shape.
Figure 3.4a shows the components of the normalized Stokes vector on the vertical axis versus the elapsed time since the start of the measurement in hours on the horizontal axis. Measurements of the polarization after the fiber are taken every $10\,$s for $48\,$h. At $0h$, so at the start of the measurement, the polarization at the input was chosen such that the output polarization was horizontally polarized. In Stokes vector formalism, this corresponds to the following vector

$$\vec{s}_H = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} ,$$

which matches the data. In the next $48\,$h, the polarization starts to randomly fluctuate due to changes in the environment of the fiber. In [34], the authors correlate these drifts with local wind speeds. This is possible since their fibers are spanned over telephone poles, where wind introduces mechanical stress. For the fibers of this thesis, no correlations such as wind or outside temperature can be identified. Since the fibers are quite long with about $23.7\,$km length and run alongside the subway, the drifts are probably an accumulation of many small localized effects such as local temperature fluctuations or local mechanical stress exhibited by construction sites. Moreover, sudden changes in the polarization are observed at around $16\,$h, $19\,$h and $43\,$h after the start of the measurement. These sudden changes are most likely not attributed to the outside environment, but somebody touching the fiber on accident inside the system - meaning in the laboratory.




(a) Red fiber




(b) Green fiber

Figure 3.4: Plot showing the transmitted polarization of the red and green fiber over time. On the horizontal axis, the time passed since the start of the measurement in hours is displayed while on the vertical axis, the three different components of the normalized stokes vector can be found. Both plots show overall great stability, with the red fiber showing more sudden changes that are most likely not related to the outside, but the lab environment such as somebody accidentally touching the fiber.

In the following, a criterion is defined that if violated, necessitates the correction of the fiber's polarization drifts. Since the quantum information after the fiber should be the same, one requires that the fidelity stays above the threshold

$$F \geq 0.99 \tag{3.10}$$

at all times. It is also technologically possible to stay above this threshold since the polarization controller of LMU compensates the polarization drifts such that fidelities of $F = 0.999$ can be achieved. For more information, see section 4.1.

To find out the earliest time this criterion is violated, the data behind the plots from figure 3.4 is used. The Stokes vectors that are measured using classical light are converted into Bloch vectors of a polarization qubit, as the Stokes vector of classical light on the Poincare sphere and a Bloch vector of a qubit on the Bloch sphere are analogous concepts [35]. Representing the Bloch vector in spherical coordinates and assuming a pure quantum state, the following equation holds true

$$\begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{bmatrix}, \tag{3.11}$$

where $s_i$ is the i-th component of the normalized Stokes vector describing the classical light and $\theta$ and $\phi$ are the angles that describe the quantum state on the Bloch sphere. Since the Stokes vectors are measured in cartesian coordinates and the Bloch vectors are represented using spherical coordinates, the following transformation rules apply between the two:

$$\theta = \arccos\left(\frac{s_3}{\sqrt{s_1^2+s_2^2+s_3^2}}\right) \tag{3.12}$$

$$\phi = \operatorname{sgn}(s_2)\arccos\left(\frac{s_1}{\sqrt{s_1^2+s_2^2}}\right) \tag{3.13}$$

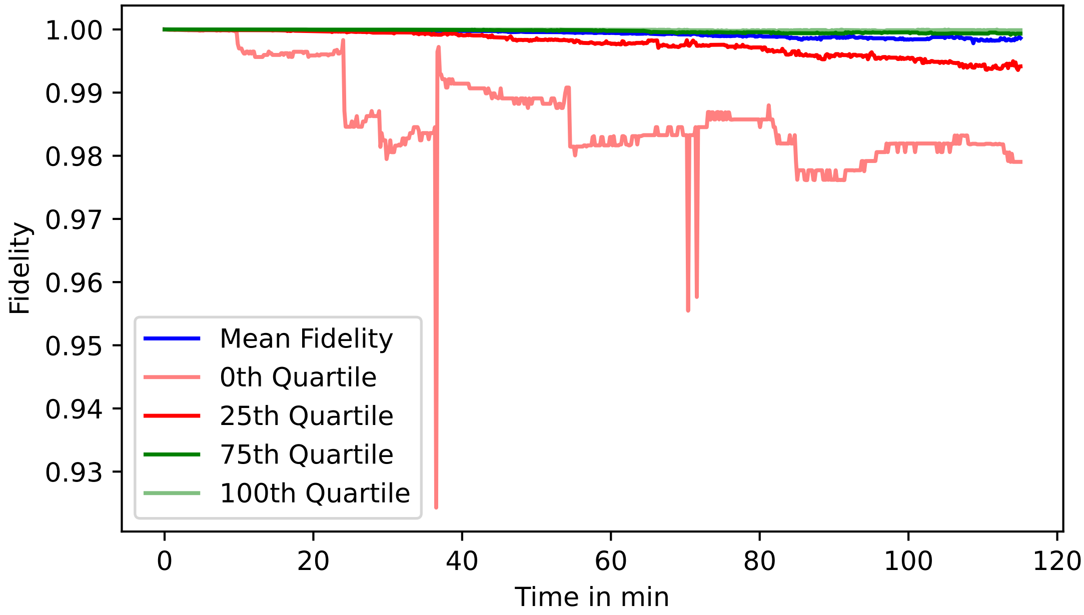
$$r = \sqrt{s_1^2+s_2^2+s_3^2} = 1 \tag{3.14}$$

Moreover, the Bloch vector corresponds to the following state in Dirac notation:

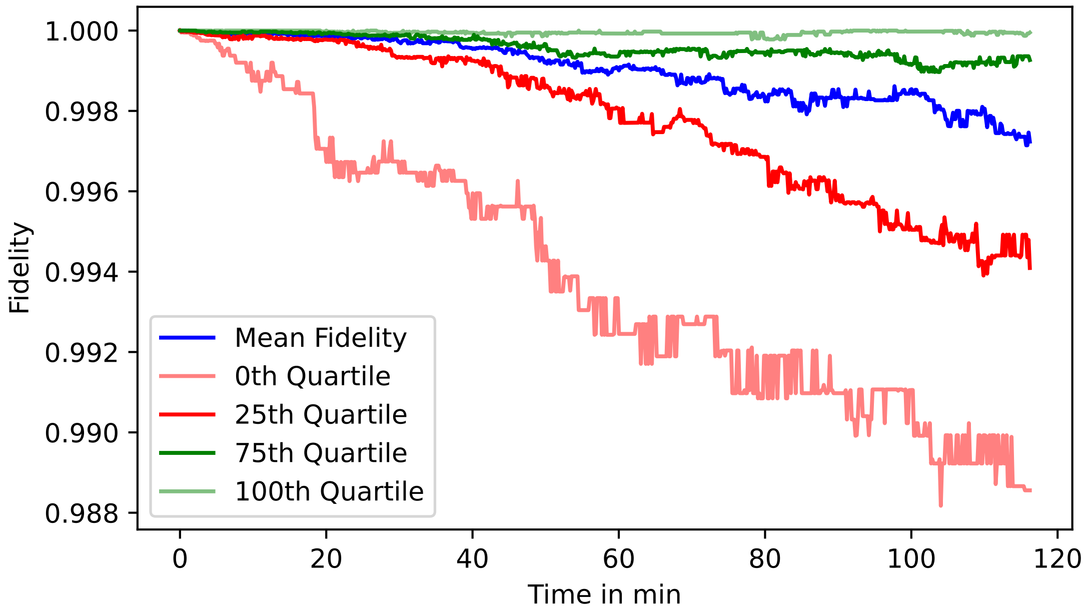
$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|\mathrm{R}\rangle + \sin\left(\frac{\theta}{2}\right)e^{i\phi}|\mathrm{L}\rangle \tag{3.15}$$

The output state at a later point in time and its associated variables are denoted with a dash and have the same structure as 3.15. Then, the fidelity between the output state $|\psi\rangle$ 3.15 at the beginning of the measurement and the output state $|\psi\rangle'$ at a later time point in time is given by

$$F = \cos\frac{\theta}{2}\cos\frac{\theta'}{2} + \sin\frac{\theta}{2}\sin\frac{\theta'}{2}e^{i(\phi'-\phi)}. \tag{3.16}$$




(a) Red fiber showing passive polarization stability of roughly $25$ min.




(b) Green fiber showing passive polarization stability of roughly $100$ min.

Figure 3.5: Plot showing the fidelity of the sent quantum state and the received quantum state depending on the time in min of the red and green fiber. The figures are created by taking figure 3.4, slicing it into $2h$ time intervals, calculating the fidelity over time in each interval and plotting the average in blue. The other colours show the $x-$th quartile fidelity at a certain point in time. For example light red, which presents the 0th-quartile fidelity, is the lowest measured fidelity, meaning all measured fidelities were found on or above this curve.

Using equation 3.16 and the data from figure 3.4, the fidelity drift for qubits in time can be calculated from the polarization drift. To do so, each time trace is cut into slices of $2$ hours - totaling 24 time traces. Then, the first data point of all of these slices is used as the initial output state $|\psi\rangle$ while the following data points are used as the state $|\psi\rangle'$ to calculate the fidelity drift for each respective slice. These time slices are then analysed using various statistical methods, which are described below.

The figures 3.5a and 3.5b show the fidelity drift in time of the red and green fiber respectively. In the following, figure 3.5b is explained in as the findings can be transferred to figure 3.5a. The plot shows different fidelity on the vertical axis versus time on the horizontal axis. The blue curve is the average fidelity from all $2$ h time slices, which is also why the horizontal axis covers time between $0$ h and $2$ h. The $25$-th quartile curve in dark red indicates the fidelity, which is larger than $25$ % of fidelities out all time slices. Therefore $75$ % of measurements are found above this curve. Similarly the $0$-th quartile in light red, which is larger than $0$ % of fidelities out of all time slices - meaning it is the lowest measured fidelity. The $100$-th quartile in light green indicates the fidelity, which is larger than $100$ % of fidelities out all time slices - meaning it is the highest measured fidelity.

The $100$-th quartile curve stays at approximately $F = 1$. This means that the best fidelity at every instance in time is close to optimal. The average fidelity, shown in blue, is seen to gradually decrease to just below $F = 0.998$ after $2$ h staying above the defined threshold $F \geq 0.99$ from equation 3.10. However, taking a look at the $0$-th quartile curve, so the worst performing fidelity at every instance in time, it drops below the threshold at roughly $100$ min. Therefore, the passive polarization stability of the green fiber can be claimed to be roughly $100$ min. Looking at the red fiber, this the passive polarization stability is roughly $25$ min. The stability is found to be lower for the red fiber since 3 dips can be seen in figure 3.5a for the worst performing fidelity in light red. As mentioned earlier for the polarization drift in figure 3.4a, this is most likely an experimenter accidentally touching the fiber and changing its polarization in the process.

This analysis shows that the fibers in this thesis are highly passively stable in comparison to other fibers. For example the fidelity in the Saarland fiber link [33] decreases below the threshold on the order of only a few seconds in the worst case scenario. However, the comparison is limited, as they use another measure for the polarization drifts, namely the process fidelity of the fiber. This is a measure of how close the operation implemented by the fiber is to the identity operation.

In the experiment, the fiber used as the quantum channel is actively stabilized every $7$ min instead of the $100$ min it is passively stable for. This is because, the above measurements are taken for a certain time of the year as well as environment. Over long timescales, the environment of the fibers can change drastically, for example if a construction site is opened around them, which would decrease passive stability. Moreover, the gradient descent algorithm that corrects the polarization drifts works best when the current polarization is not far from the optimal values, which is why a smaller time interval than required is used for active stabilization.

## 3.5 Summary and Literature Review

In this chapter, the optical telecom fibers were characterized for their use in quantum networks. Fundamentally, the fibers run from LMU in the city centre of Munich along the U6 train line to Garching Forschungszentrum, where they end at MPQ. Using an optical time-domain reflectometer, the green fiber was measured to be $23.71$ km and the red fiber was measured to be $23.74$ km long. Additionally, by using the same measurement, the fibers were determined to have the same splice positions and did not show bigger faults like breaks in the fiber core.

Afterwards, two calibrated power meters were used to measure the transmission through each fiber, where an attenuation at the desired wavelength of $1514$ nm of $8.41$ dB for the green and of $9.14$ dB for the red fiber is found. Moreover, the polarization dependent losses were measured to be $\mathrm{PDL} = (0.109 \pm 0.019)$ dB for the green and $\mathrm{PDL} = (0.083 \pm 0.024)\,\mathrm{dB}$ for the red fiber.

The green fiber was chosen as the quantum channel due to its lower attenuation, which is crucial when dealing with transmission of quantum information, as it cannot be cloned or amplified.

| Fiber | $\lambda$ in nm | $l$ in km | Loss in dB/km | PDL in dB | Stability |
|---|---|---|---|---|---|
| Green Fiber (this work) | $1514$ | $23.71$ | $0.355$ | $0.109(19)$ | $\approx 100$ min |
| Red Fiber (this work) | $1514$ | $23.74$ | $0.385$ | $0.083(24)$ | $\approx 25$ min |
| Saarland Link-Q [33] | $1550$ | $14.35$ | $0.725$ | $0.08(9)$ | $\sim 1$ min[1] |
| Saarland Link-C [33] | $1550$ | $14.35$ | $0.62$ | $0.08(9)$ | $\sim 1$ min |
| BARQNET A [34] | $1550$ & $1350$ | $42.5$ | $0.28$ & $0.391$ | ? | $\sim 100\,\mu$s[2] |
| BARQNET B [34] | $1550$ & $1350$ | $42.5$ | $0.4$ & $0.515$ | ? | $\sim 100\,\mu$s |
| Fiber Delft [36] | $1588$ | $15$ | $0.373$ | ? | $\sim 1$ s[3] |
| Fiber The Hague [36] | $1588$ | $10$ | $0.52$ | ? | $\sim 1$ s |
| Toshiba Fiber [37] | $1320$ | $18.23$ | $0.642$ | ? | $\sim 1$ min [4] |
| Hefei A - Server [38] | $1342$ | $10.1$ | $0.36$ | ? | $\sim 1$ h [5] |
| Hefei B - Server [38] | $1342$ | $9.6$ | $0.48$ | ? | $\sim 1$ h |
| Hefei C - Server [38] | $1342$ | $11.5$ | $0.46$ | ? | $\sim 1$ h |

Table 3.2: Table comparing different field deployed fibers that are used for quantum networks and links. They are compared with regards to their functionality using polarization qubits - although [34] works with time-bin qubits and [36] as well as [38] work with number state qubits. The first column compares the intended wavelength of light $\lambda$ to be sent through these fibers, while the second column shows the length $l$ of each fiber. The third and fourth column compare the attenuation and polarization dependent losses (PDL) respectively. The stability in the last column is the scale of time on which the different authors need to correct for polarization drifts in their fiber. The exact stability criteria when to stabilize the fiber differ between these works - see footnotes or text for details.

A comprehensive literature review comparing different field deployed fiber networks with regards to their functionality using polarization qubits is included in table 3.2. The additional fiber networks are located in Saarland, Germany [33], in Boston, United States [34], in between Delft and Den Hague, Netherlands [36] in Cambridge, United Kingdom [37] and in Hefei, China [38].

In the comparison, the fibers investigated in this theses perform well, as they have lower general losses per kilometer at around $0.35\,\frac{\mathrm{dB}}{\mathrm{km}}$ with very high passive polarization stability on the order of $10$ min, where most fibers networks show substantially higher attenuation and lower stability except the fiber network in Hefei. One of the main reasons, why these fibers perform more poorly in terms of stability is that for example in [33] and [34], the fibers are spanned over telephone poles, making them highly susceptible to environmental changes such as rain and wind.

However, one needs to keep in mind that the criteria on which polarization stability was evaluated differs between each of these works and thus comparisons may only be accurate up to orders of magnitude. In the following, these different criteria are briefly presented: The fibers presented in this thesis are defined to be stable until the fidelity between input and output qubit to the fiber drops below $F = 0.99$ in the worst possible case, see equation 3.10 for a detailed discussion of this criterion. In [33], it is the time until the process fidelity of the quantum map of the fiber drops below $F = 0.99$, which is estimated from figure 5 in [33]. In [34], the authors mention that they would need a $\sim 10$ kHz-stabilization system to achieve high fidelity polarization stability. The authors in [36] claim they need to stabilize on the order of seconds in order to achieve polarization alignment within a few percent. In [37], the authors claim to correct their fiber once the fidelity drops below a threshold of $98.5$ % and every $11$ min, comparable to the threshold used in this thesis. The authors from [38] responded to an inquiry regarding their paper, saying that they check every $26$ ms and correct the polarization once the photon number on a superconducting nanowire single photon detector (SNSPD) is above a certain threshold, which they say typically happens on a timescale of several hours.

[1] The time until the process fidelity of the fiber drops below $0.99$ is estimated from figure 5 in [33].

[2] The authors discuss they need $\sim 10-$kHz stabilization in order to have high fidelity polarization stabilization.

[3] Authors find that a feedback system of a few Hz aligns the polarization within a few percent.

[4] The authors correct the fiber once the fidelity drops below $98.5$ % and every $11$ min.

[5] The authors responded to an inquiry about their paper that their typical drift time scale is several hours.

# Chapter 4

# Atom-Photon Entanglement

Using the optical telecom fibers characterized in the previous chapter 3, LMU now distributes photons that are entangled with their atom to MPQ as a first step towards atom-atom entanglement.
Therefore, this chapter is devoted to establishing and analysing the entanglement of the resulting atom-photon state between LMU and MPQ using the theoretical tools described in chapter 2.

As motivated at the start of chapter 3, the photon emitted by LMU's $^{87}\mathrm{Rb}$-node needs to be converted from near-infrared to telecom frequencies using a quantum frequency converter for lossless propagation. After arriving at MPQ, these photons are converted back to near-infrared frequencies so that they can be stored into the $^{87}\mathrm{Rb}$-node at MPQ. Therefore, the atom-photon entangled state will be analysed with a memory-compatible photon with a wavelength of about $780\,\mathrm{nm}$. If entanglement is confirmed, the next step would be to store this incoming photon into the $^{87}\mathrm{Rb}$-node at MPQ. By choosing the right atomic excitation scheme, atom-atom entanglement can be created, which would complete the construction of the quantum link.

This chapter starts off by giving an overview of the experimental devices in section 4.1 and their particular function within the experiment. Specifically, the filter cavity, which filters noise coming from the quantum frequency converter, and the detection setup that were built within the framework of this thesis are examined in more detail. Afterwards, a theoretical model of the experiment is presented in section 4.2 that predicts the physical quantities of interest such as the entangling rate and fidelity of the shared atom-photon state in each entanglement attempt. Next up, measurements of the entangling rate and fidelity are analysed in section 4.3 and compared to the theoretical expectations from the preceding section 4.2. Finally, the main findings of this chapter are summarized in section 4.4 and compared to other experiments.

## 4.1 Setup

A schematic of the setup showing critical components needed to measure either atom-photon or atom-atom entanglement between LMU and MPQ can be found in figure 4.1. Components deemed less critical are omitted to keep the schematic clear.
The setup is described starting at LMU and ending at MPQ - meaning describing the components from left to right.

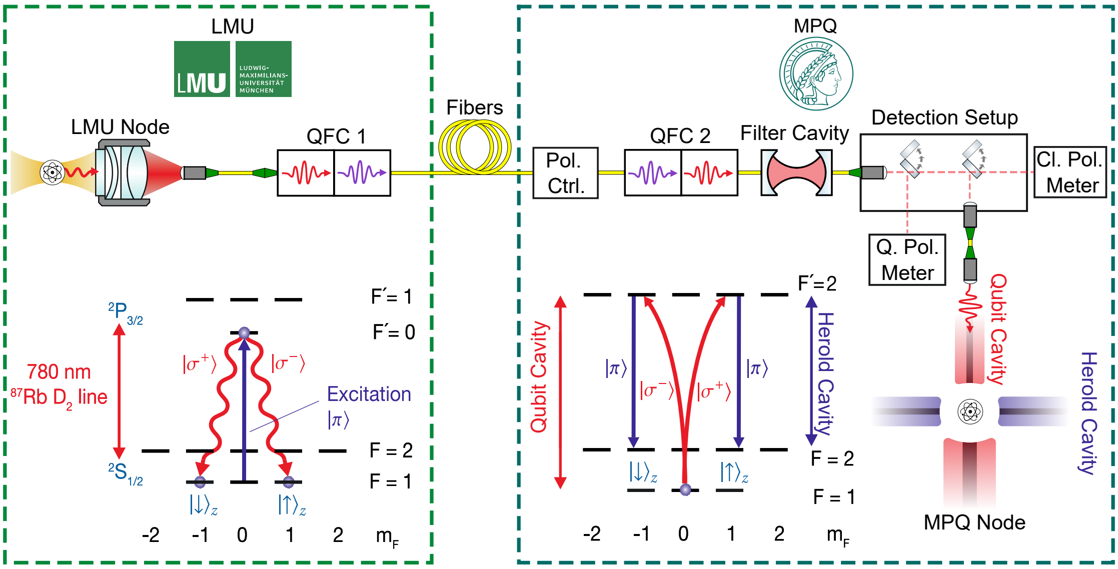



Figure 4.1: Schematic of the atom-photon or -atom entanglement setup. From left to right: Atom sitting in an optical dipole trap (LMU node), emitting photons entangled in polarization with the atomic spin, as shown in the excitation scheme below, and which are collected by an objective into fiber. Quantum frequency converter at LMU (QFC 1) used to convert from infrared to telecom frequencies. Long fiber used as the quantum channel between LMU and MPQ, which was characterized in chapter 3. Polarization controller (Pol. Ctrl.) to correct fiber polarization drifts. Second quantum frequency converter (QFC 2) at MPQ for converting back to infrared frequencies. Filter cavity setup for filtering noise, mostly originating from QFC 2. Detection setup managing the kind of experiment being performed, e.g. atom-photon entanglement by routing single photons to a polarization analysing stage (Q. Pol. Meter) or atom-atom entanglement by routing the single photons towards MPQ node. It also routes classical light to the classical polarimeter (Cl. Pol. Meter) in certain time intervals which measures the polarization drifts of the fiber. The atomic level scheme at MPQ shows the relevant transitions for storing a photon in the atom-atom entanglement experiment.

The first key component is the node at LMU, which can be seen on the very left in figure 4.1. The node consists of an optical dipole trap, which traps a single $^{87}\mathrm{Rb}$-atom. These atoms are cooled and then prepared in the $5^2\mathrm{S}_{1/2}$-state with $F = 1, m_f = 0$ from which they are excited towards the $5^2\mathrm{P}_{3/2}$-state with $F' = 0, m_{f'} = 0$ using a $\pi-$polarized Gaussian laser pulse in time. In addition with the following free space decay, the photon obtains a certain temporal and spectral shape. The spectra of different devices, particularly of the filter cavity described in section 4.1.1, must be matched to the photon's spectrum to ensure its transmission through them. Due to the dipole selection rules of free space decay, where only decay into $5^2\mathrm{S}_{1/2}$-state with $F = 1$ manifold is possible, the composite atom-photon state becomes:

$$|\phi^+\rangle = \sqrt{\frac{1}{3}}\Big(|m_f = -1, m_s = +1\rangle + |m_f = +1, m_s = -1\rangle + |m_f = 0, m_s = 0\rangle\Big).$$

The quantization axis of the system is defined to line up with the optical axis of the lens that collects the photons into the fiber. Since $\pi$-polarized photons ($m_s = 0$) are emitted orthogonal to this axis by definition, they are not coupled into the fiber. Hence, after the fiber the maximally entangled state from equation 2.14 is obtained. The excitation scheme that was just described is shown in the bottom left of figure 4.1 and more information about it can be found in [39].

The next critical component is the first quantum frequency converter, which converts the entangled photons with a wavelength of about $780$ nm, as given by the atomic excitation scheme at LMU, to $1514$ nm. This is done to achieve higher transmission through the optical fibers with a length of about $23.7$ km of roughly five orders of magnitude. A wavelength of $1514$ nm is chosen instead of some other telecom wavelength, since the noise generated by conversion to this wavelength is very low. The quantum frequency converter works by using a nonlinear crystal that is pumped with light at $1610$ nm to achieve conversion from light at $780$ nm to $1514$ nm or vice versa by difference or sum frequency generation respectively. Moreover, it is called a "quantum" frequency converter, as it converts the frequency of a single photon while maintaining its quantum state [40]. Maintaining the quantum state after conversion is essential, as the quantum information is encoded in the polarization state of a single photon. For more information on quantum frequency converters, the reader is referred to [41], where the working principle as well as similar atom-photon entanglement distribution experiments using a quantum frequency converter are explained.

The next components are the roughly $23.7$ km long optical fibers connecting LMU and MPQ, where one is used to exchange classical information while the other is used as the quantum channel to distribute the entangled photons from LMU to MPQ. For more information, especially regarding the characterization of the quantum channel, see chapter 3.

After the optical fibers, the single photons reach MPQ. Here they are first incident on a polarization controller as can be seen in 4.1. The polarization controller works by inducing mechanical stress on the fiber, which is used to actively compensate polarization drifts in the fiber and any other component in between sender and polarization reference. To measure these drifts, LMU sends classical light instead of entangled photons over the same optical path in seven minute intervals during the experiment. It is important to highlight that the compensation only works if both classical light and entangled photons take the same optical path such that they are subject to the same polarization drifts. The light is then measured using the classical polarimeter, which is discussed in more detail in section 4.1.2. These measurements are then used to produce an error function, which is minimized using a gradient descent algorithm that controls the polarization controller.

Next comes the second quantum frequency converter at MPQ, which converts the light from $1514$ nm back to $780$ nm. More importantly, the pump laser is tuned to a slightly different frequency compared to the first quantum frequency converter, as the utilized atomic transitions differ between LMU and MPQ. At LMU, transitions between the hyperfine manifolds $F = 1$ and $F' = 0$ are used for photon emission, while at MPQ transitions between the $F = 1$ and $F' = 2$ manifold are used to store the photon. The atomic transitions at LMU can be seen in the bottom left of figure 4.1, while the transitions at MPQ can be seen on the bottom right of figure 4.1.
This demonstrates another strength of the quantum link investigated in this thesis, namely that different atomic excitation or storage transitions as well as in principle entirely different hardware platforms can be connected to each other by adjusting the quantum frequency converters.

Since both quantum frequency converters produce noise, a filter cavity setup was built that acts as a narrow band-pass filter for photons at the desired wavelength of roughly $780$ nm. It is placed directly after the second quantum frequency converter. The section 4.1.1 has all the relevant details about the setup as well as its filtering capabilities.

Last but not least, the light from LMU is incident on the detection setup, where the experimenter can choose which kind of detection is used for the incoming light. If single photons are incident on the detection setup, they are guided towards a polarization analysis stage called quantum polarimeter. If however, classical light is incident on the detection setup, the light is guided towards a classical polarization analysis stage called classical polarimeter. Moreover, for future atom-atom experiments, the light can also be guided towards single atoms for storage. More information regarding the detection setup, can be found in 4.1.2.

### 4.1.1 Filter Cavity Setup

In the following, the filter cavity setup is examined in more detail, as it was built specifically within the framework of this thesis to filter the noise coming from the quantum frequency converters. The amount of noise from the quantum frequency converters is roughly constant with frequency within small frequency ranges, such as the ranges of interest here.

#### Working Principle

The basic idea of how to make a filter out of a cavity, is to tune one of the cavities many possible resonance frequencies into resonance with the qubit's frequency by controlling the cavity's length such that the qubit is transmitted. By additionally using a band-pass filter behind the cavity, such as a Volume Bragg grating (VBG), only the qubit's resonance frequency can be reflected while the other cavity resonances are transmitted. This effect can be used to filter out noisy light at frequencies different from that of the qubit. A VBG works by alternating two materials with different refractive indices that provide different interference criteria in front of the VBG depending on the wavelength of light. Thus incoming light is either reflected, in the case of constructive interference in front of the VBG or transmitted, in the case of destructive interference.

### Design and Setup

In the following, the setup is explained first using the schematic representation in figure 4.2. Secondly, certain design parameters of devices appearing in the filter cavity setup are elaborated on.

There are two beam paths to consider in the filter cavity setup: The beam path in red is taken by the entangled photons that are sent by LMU, which were just converted by a quantum frequency converter. First, they are emitted into free space from a fiber coupler, transmitted through the cavity and reflected off the VBG. Then, they are coupled back into fiber and sent towards the detection setup for analysis. A flip mirror can be flipped into the beam to reflect it, where the reflected beam is shown with a dashed line, towards a photodiode (PD). When the flip mirror is flipped in, classical light at the qubit's frequency can be used to determine the mode matching to the cavity. To do so, the cavity's length must be scanned - one such measurement can be seen in figure 4.4. The other beam path in blue is used to lock the filter cavity with the Pound-Drever-Hall technique to a certain length and thus resonance frequency. To do so, a laser at a wavelength of $795$ nm is emitted through a third fiber coupler with a power of about $56.5\,\mu$W seen in the top right of figure 4.2. Its light is first incident $\lambda/2$-waveplate. This is used to maximize the transmission through the PBS afterwards that splits the light in horizontal and vertical polarization. After the PBS, the light is incident on a $\lambda/4$-waveplate, which sets the phase of the light such that the back reflected light from the cavity is guided towards the other photodiode in the setup. Finally, the reflected light is detected using this photodiode.
One component that is not described yet, are the narrow-band filters that are placed in front of every fiber coupler. In the red beam path, these transmit light at the wavelength of the qubits at around $780$ nm and are used to filter the classical lock light at $795$ nm. This only works if the laser only emits at its peak wavelength of $795$ nm. However, there is also a certain spectral overlap with $780$ nm. Therefore, narrow-band filters only transmitting at around $795$ nm are placed into the beam path of the lock light. Using this setup of filters, the lock light that is transmitted to the detection setup and finally detected on the SNSPDs is reduced from $750$ kHz, which is the count rate in saturation, down to a combined $1.11$ Hz.
So far, the optics in figure 4.2 have been described in detail. Next, the different electrical control signals and devices that are needed to lock the cavity using the Pound-Drever-Hall technique are explained. The electric signal of the photodiode detecting the lock light is forwarded to the input of a Bias T, which separates alternating current (AC) and direct current (DC) parts of the signal. The AC part of the signal is then used by a Red Pitaya, which runs a community made program for a proportional–integral–derivative (PID) controller [42]. To use the PID controller to lock the cavity, the lock light is modulated using an electro-optic

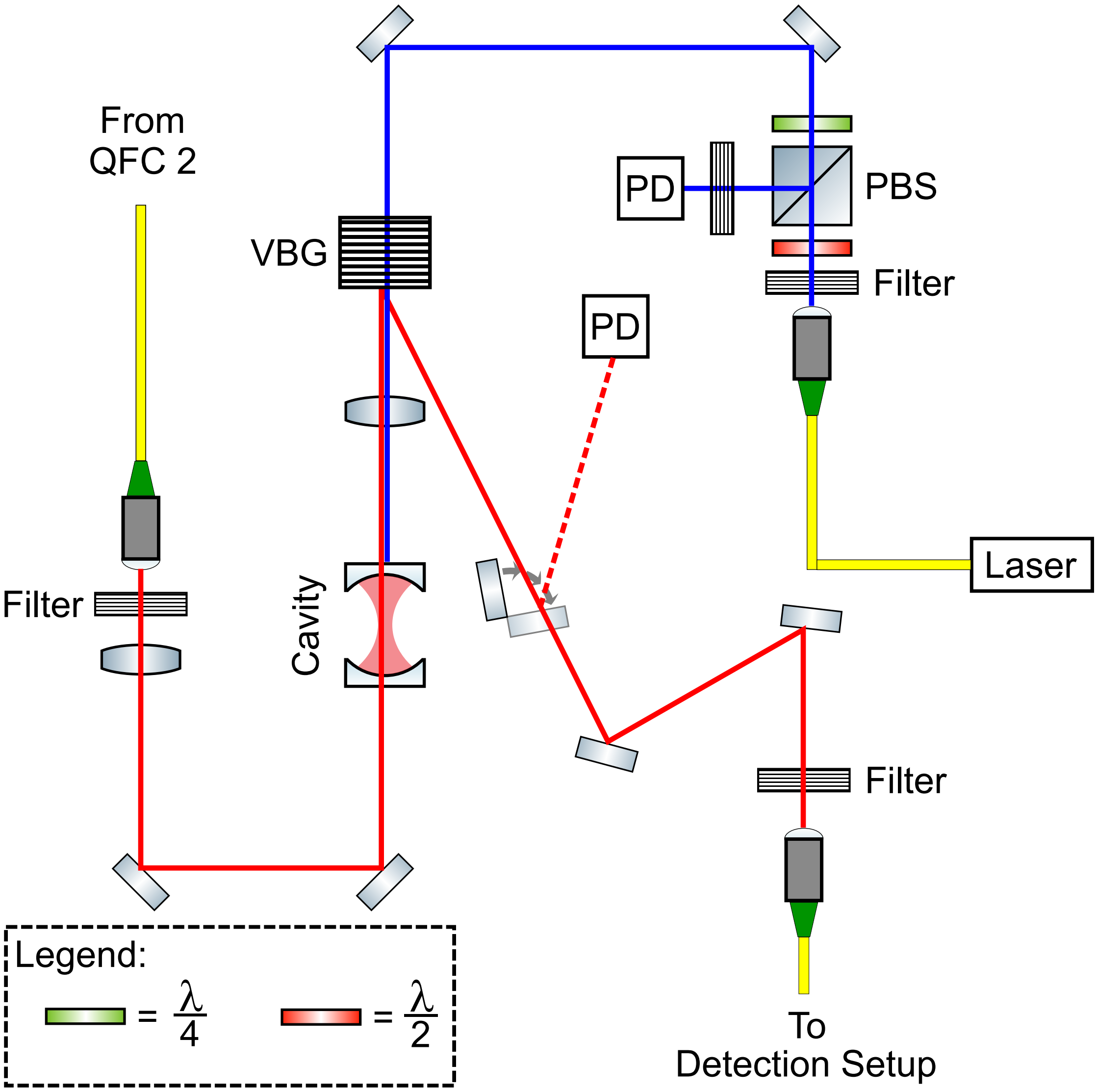



Figure 4.2: Figure showing the filter cavity setup. On the middle left sits the fiber coupler out of which the single photons from the LMU node arrive. Following the red beam path, noise is suppressed by passing through a cavity and subsequent reflection from a Volume-Bragg grating (VBG). Then, the light is coupled back into fiber leading to the detection setup. The blue beam path is taken by the light at $795$ nm used to lock the cavity to a certain frequency. It is reflected of the cavity and finally collected on a photodiode (PD), whose signal is used for locking the cavity with the Pound-Drever-Hall technique.

modulator controlled by the Red Pitaya such that its reflection resembles the error signal found in [43], which explains the Pound-Drever-Hall locking technique. This technique does not work however, if the lock light itself is not at a stable frequency as this is the reference frequency, which the PID locking mechanism will stabilize the cavity to. Therefore, the lock laser is stabilized by interfering it with a stable frequency comb reference and using this signal as feedback to the laser to lock it using a PID controller.

Before the setup in figure 4.2 could be built, the interplay of parameters from different devices such as lenses, fiber couplers, VBG and cavity had to be carefully designed to maximize both noise filtering and transmission of signal light. The different design parameters are discussed below:
The cavity was already designed and built by LMU and detailed information can be found in the Bachelor's thesis of Wanlin Li [44]. The most important design features are summarized in the list below:

- $\mathrm{FSR} = 18.2$ GHz ($\mathrm{FSR}$ is the free spectral range)
- $\mathrm{FWHM} = 29.9$ MHz ($\mathrm{FWHM}$ is the full-width at half maximum)
- beam waist $w_0 = 78\,\mu$m

To match the incoming spatial Gaussian beam's waist width to the beam waist $w_0 = 78\,\mu$m of the fundamental cavity mode, the parameters of the different lenses are optimized for mode matching using a program called "GaussianBeam". The following parameters are found to be optimal:

- Distance between lenses and center of cavity: $250$ mm
- Lenses focal lengths: $250$ mm
- Fiber coupler focal length in qubit path: $8$ mm
- Fiber coupler focal length in lock path: $12$ mm

These are chosen such that the overlap between fundamental cavity mode and incoming qubit beam is $99.99\,\%$, which allows for high transmission of the signal. Moreover, this mode matching is achieved when the incoming beam to the lens is collimated and the distance between a lens to the center of the cavity is $250$ mm as mentioned above.
The VBG is characterized by the manufacturer OptiGrate with the following parameters:

- Input angle for designed wavelength: $2 \pm 1\,^\circ$
- Output angle for designed wavelength: $-8 \pm 1\,^\circ$
- Spectral bandwidth: $< 50$ GHz

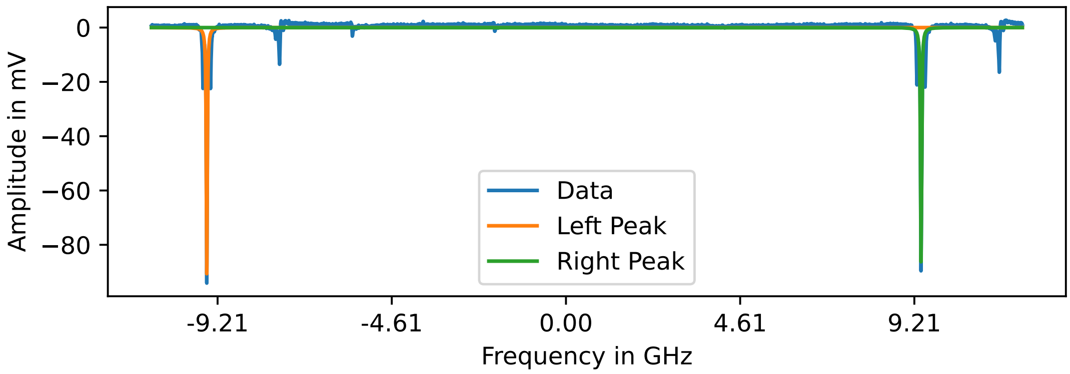
Since the cavity's free spectral range fits two times into the spectral bandwidth of the VBG, up to three cavity modes can be reflected of the VBG in the worst case instead of only one at the desired frequency.
Each resonance of the cavity has approximately the same amount of noise associated to it, since it can be assumed that the noise generated by the quantum frequency is constant with frequency in such a small frequency range. Therefore, choosing a VBG with a smaller spectral bandwidth is predicted to reduce the noise by a factor of $1/3$ by cutting out the additional two resonances that only carry noise. However, since there is no VBG available

with smaller spectral bandwidth, reducing the length of the cavity to increase its $\mathrm{FSR}$ and cut out the additional two resonances this way can be investigated to mitigate the noise. By reducing the length the cavity, also its $\mathrm{FWHM}$ would be increased. Since the noise can be modelled to depend linearly on the $\mathrm{FWHM}$, see equation 4.7, this would increase the noise again. Thus it remains to be seen whether reducing the length of the cavity works to mitigate the noise. Another option would be to change to another cavity entirely with different finesse, which is the ratio of $\mathrm{FSR}$ to $\mathrm{FWHM}$.

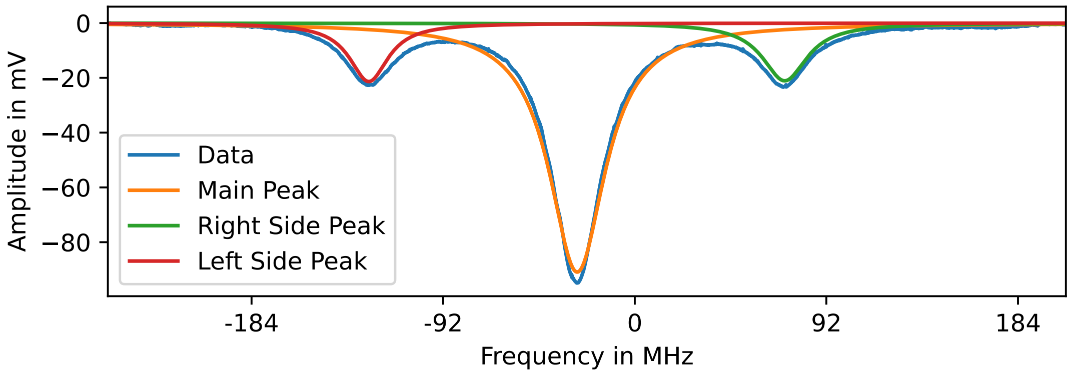
### Characterization

After explaining the idea and design of the filter cavity in the previous sections, some characteristic measurement of the cavity are presented in the following.
As already mentioned, the filter cavity was built by LMU [44] and transported to MPQ. Therefore, measurements of the free spectral range and full width at half maximum, are repeated to check whether the cavity was altered or damaged during transportation.

To check both the $\mathrm{FSR}$ and $\mathrm{FWHM}$, the length of the filter cavity is scanned by applying a triangular voltage signal to the piezoelectric motor and collecting the reflection of the lock light from the cavity. Moreover, this light is modulated before reflection with a sinusoidal frequency of $100$ MHz to get sum and difference frequency signals using a fiber electro-optic modulator. This modulation helps to convert between time and frequency domain, where the conversion yields information about frequency differences. The reflected lock light is detected using a photodiode and recorded in time using an oscilloscope triggered to the triangular voltage signal controlling the length of the cavity. Using the conversion factor $c$, which is discussed below, this data is converted to the frequency domain and plotted in figure 4.3.
The side peaks due to sum and difference frequency generation from modulating the carrier frequency as well as the carrier frequency itself are clearly visible in figure 4.3b. By fitting the side peaks with a Lorentzian in the time domain, their positions are found to be $t_l = 48.52$ ms and $t_r = 48.95$ ms. The aforementioned numbers are given without uncertainty, as the fit's uncertainty is negligible. Moreover, systematic uncertainties such as the measurement precision are not accounted for either. Using these numbers, a conversion factor between scanning time and corresponding resonance frequency of light can now be calculated. Since the side peaks should be spaced $\Delta f = 200$ MHz apart due to the modulation, a conversion factor of $c = \frac{\Delta f}{\Delta t} = 460.72\,\frac{\mathrm{MHz}}{\mathrm{ms}}$ is found, where $\Delta t = t_r - t_l$.
Fitting the carrier peak in figure 4.3b with a Lorentzian in the time domain, the full width at half maximum is given by $\mathrm{FWHM} = 0.071$ ms. This corresponds to $\mathrm{FWHM} = 32.79$ MHz by using the conversion factor $c$, which roughly matches with $\mathrm{FWHM} = 29.9$ MHz as measured in [44]. Fitting both recorded fundamental resonances in figure 4.3a with a Lorentzian in the time domain to get the carrier peak's timing, the following timings are obtained: $t_l^0 = 49.38$ ms and $t_r^0 = 90.38$ ms. By taking their difference and converting the result using $c$, one acquires $\mathrm{FSR} = 18.9$ GHz, which matches the theoretical design value of $\mathrm{FSR} = 18.2$ GHz [44].




(a) By fitting both peaks with a Lorentzian, and taking the difference of the center frequencies, the $\mathrm{FSR}$ is found to be $\mathrm{FSR} = 18.9$ GHz. This value matches with it's theoretical design value of $\mathrm{FSR} = 18.2$ GHz as noted in [44].




(b) Zoom in on the left from figure 4.3a. The sum and difference as well as carrier frequency peaks are clearly visible. By fitting the carrier peak with a Lorentzian, a $\mathrm{FWHM}$ of $\mathrm{FWHM} = 32.79$ MHz is found, which matches with $\mathrm{FWHM} = 29.9$ MHz as measured in [44]. Moreover, this figure in the time domain is used to calculate the conversion factor of $c = \frac{\Delta f}{\Delta t} = 460.72\,\frac{\mathrm{MHz}}{\mathrm{ms}}$.

Figure 4.3: Plots showing the measurements used to characterize the filter cavity. The amplitude of light in mV as measured by a photodiode is on the vertical axis. On the horizontal axis, one finds the frequency difference to a common reference point in GHz or MHz respectively. The data for these plots was taken by scanning the length of the filter cavity and reflecting light that was modulated with side bands from the cavity. This reflected light is detected with a photodiode and recorded in time using an oscilloscope. The modulation is used to convert between time and frequency domain.

In the subsequent paragraph, the mode matching between the signal light which has a Gaussian intensity profile in space to the Gaussian-like TEM00 mode of the cavity is discussed and analysed. In figure 4.4, the vertical axis shows the amplitude of the signal light

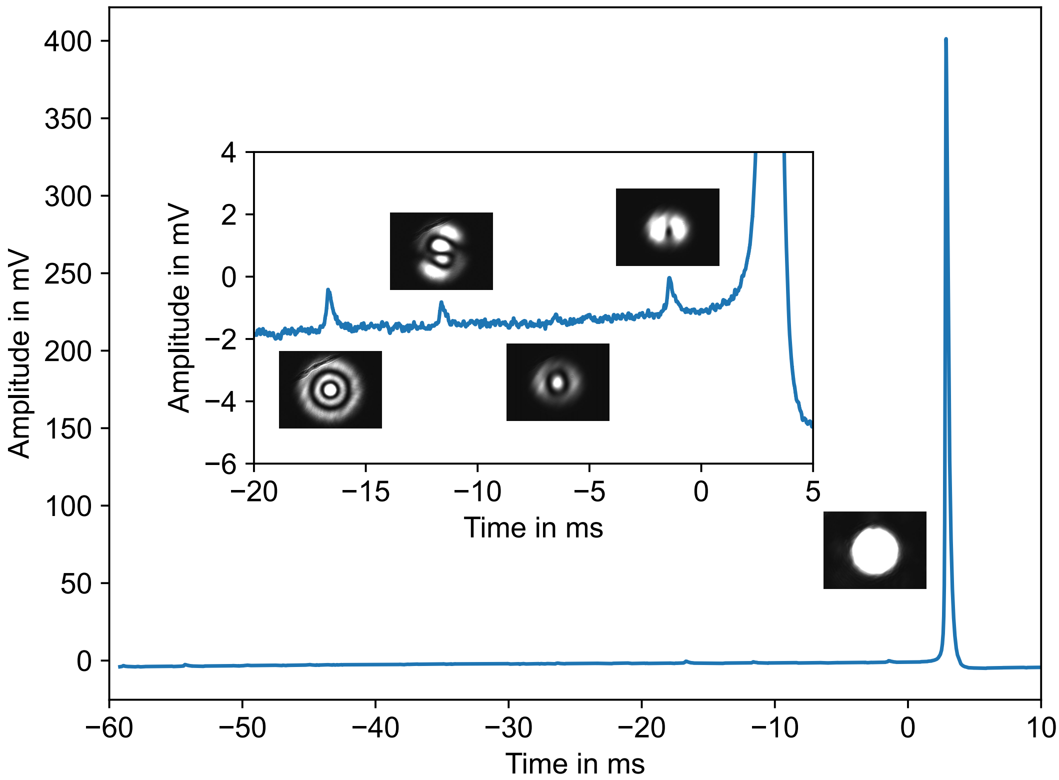



Figure 4.4: Figure showing the amplitude of the light passing through the cavity in mV as measured by a photodiode versus the time of the scan in ms, which corresponds to the cavity's length. The inset, shows a zoom-in on the small peaks between $-20$ mV and $0$ mV. The pictures show the respective cavity modes as imaged by a camera. The side mode suppression between the main TEM00 mode and the highest side peak is calculated to be $> 24$ dB, which confirms good coupling to the cavity modes.

in mV as measured by a photodiode after transmission through the cavity. The horizontal axis shows the scanning time in ms. The scanning time corresponds to the cavity's length, as a triangular voltage signal controls the length of the cavity.

By beam walking such that the light hits the center of the cavity perpendicularly and by optimizing where the lens is hit, the mode matching of the incoming beam to the Gaussian-like TEM00 mode of the cavity is optimized. The optimization is complete if the amplitude of every other mode apart from the TEM00 mode almost vanishes as can be seen in figure 4.4, which is the case if the side mode suppression is maximized. Additionally, pictures of these different modes are taken using a camera and plotted close to the corresponding peak measured on the photodiode in figure 4.4. The rightmost mode with an amplitude of $402$ mV

is the TEM00 and the leftmost mode in the inset[1] is the TEM20 mode with an amplitude of $1.3$ mV. From this a side mode suppression of $24.89$ dB can be calculated, which is defined here as the amplitude ratio of the fundamental TEM00 and the higher order mode with the next highest height. A side mode suppression value of $24.89$ dB demonstrates good suppression of higher order modes.
Supported modes by the cavity are a mixture of Gauss-Laguerre and Gauss-Hermite modes, as for example the leftmost mode is the TEM20 Gauss-Laguerre mode and the second mode from the left seems to be a TEM03 Gauss-Hermite mode. To come to the different solutions of Gauss-Laguerre and Gauss-Hermite modes, one assumes circular or rectangular symmetry for the cavity, which implies that the cavity in this thesis has a mixture of these symmetries as well.

### Performance

Finally, the performance of the filter cavity is quantified by measuring the internal efficiency, which is the transmission of light from behind the fiber input coupler to right in front of the fiber output coupler. Power measurements are collected for $10$ s for averaging using a power meter, as the power measurements fluctuate on short timescales. These power fluctuations can occur due to fluctuations in transmission through the cavity for example, which happen if the locking mechanism is not perfectly stable. The internal efficiency is measured to be $T_{in} = (91.57 \pm 0.94)\,\%$ and the fiber coupling of the output coupler is measured to be $T_{fc} = (82.06 \pm 0.96\,)\%$. Therefore, the external efficiency can be found to be $T_{ext} = (75.14 \pm 1.17)\,\%$. The aforementioned uncertainties are calculated using Gaussian error propagation. However, systematic uncertainties such as long-term power drifts between measurements are not accounted for.
The internal efficiency can be represented as follows $T_{in} = T_m^4 T_{cav} T_{VBG} T_{lens}^4$, where $T_m$ is the reflection of a mirror, $T_{cav}$ is the transmission of the cavity, $T_{VBG}$ is the reflection from the VBG and $T_{lens}$ is the transmission through the surface of a lens. For mirrors one can assume a reflectivity of $T_m = 99.5\,\%$ as stated in their specifications sheet, while a reflection of $T_{VBG} = (96.39 \pm 0.2)\,\%$ is measured for the VBG. A standard transmission value of $T_{lens} = 99.8\,\%$ is assumed for the surface of a lens. Thus by rearranging the internal efficiency formula to $T_{cav} = \frac{T_{in}}{T_m^4 T_{VBG} T_{lens}^4}$, the transmission through the cavity itself is estimated to be $T_{cav} = (97.7 \pm 1)\,\%$, where the error is calculated using Gaussian error propagation. The cavity was built similar to [41]. Here the author reaches a transmission of $T = 96.5\,\%$[41], but for light at $\lambda = 1522$ nm. Nevertheless, it can be concluded that the internal efficiency of the filter cavity setup is quite optimized.
However, one thing that still needs improvement is the fiber coupling as one could expect to achieve around $90\,\%$, while $T_{fc} = (82.06 \pm 0.96\,)\%$ is observed in the experiment. One way to increase this number would be to build a telescope in front of the fiber coupler to achieve better mode matching. Therefore, if the fiber coupling were to be increased to around $90\,\%$ with the help of a telescope for example, the external efficiency could be increased to $T_{ext} = 82\,\%$.

[1] There certainly exist higher order modes that can also be seen if one looks very carefully.

### 4.1.2 Detection Setup

The detection setup is presented in figure 4.5. It serves several functions. Most generally, it controls which kind of experiment is being performed by routing the light to three different parts of the setup:

1. Atom-photon entanglement using the quantum polarimeter
2. Atom-Atom Entanglement (atom trapping as well as analysis setup not shown here)
3. Polarization analysis using the classical polarimeter

More specifically, for all of these experiments a common reference point in polarization has to be set such that the polarization of light is the same regardless of the experiment. To do so, the light first travels through the **compensation** part of the whole detection setup, which is seen in the bottom left of figure 4.5. Here, the reference point to which all other points are compensated to, is marked as a red dot with the letter A.

The compensation procedure works as follows: At first, one of two orthogonal polarizers on the Bloch sphere is placed at reference point A. Then, a Thorlabs polarimeter is placed either at point B or C depending on which optical path should be compensated. The stack of waveplates in the beam path between the two points are then turned such that the polarization on the polarimeter and the polarizer's polarization are the same. The same is done for the second polarizer. This procedure has to be repeated until both polarizers are in agreement with the polarization measured by the polarimeter at the same time. If this is the case, all unwanted polarization shifts, such as those caused by reflection from a mirror, are compensated for ensuring that the polarization at points B and C matches the polarization at point A. This is important, since it also guarantees that the polarizations at point B and C are the same. Therefore, if LMU now compensates the fiber drifts by sending classical light to the classical polarimeter through the whole setup, the drifts are also compensated for the optical path leading to the quantum polarimeter. This ensures that the photon's polarization emitted at LMU and the one that is detected using the quantum polarimeter at MPQ are the same.

The **classical polarimeter**, seen at the top of figure 4.5, works by splitting incoming light with a 50/50 beam splitter into two different arms. Using a quarter- and half-waveplate in each arm, the basis of the polarization analysis is set to H/V and D/A respectively. Then, a polarizing beam splitter in each arm splits the aforementioned polarizations into different arms respectively, where they are detected on a photodiode. By knowing the voltage offset and sensitivity of the photodiodes as well as parameters of the 50/50 beam splitter and the polarizing beam splitters, the exact polarization of the light can be inferred.

The **quantum polarimeter** operates similarly, but it only uses one polarizing beam splitter. It can be seen in the bottom right of figure 4.5. Therefore, it can only analyse the polarization of an incoming photon in one basis, which again is set by a a quarter- and half-waveplate. Finally, these photons produce a voltage pulse once they are absorbed by a superconducting nanowire single-photon detector. The polarization is then determined depending on which detector yields the voltage pulse, also informally referred to as a "click".

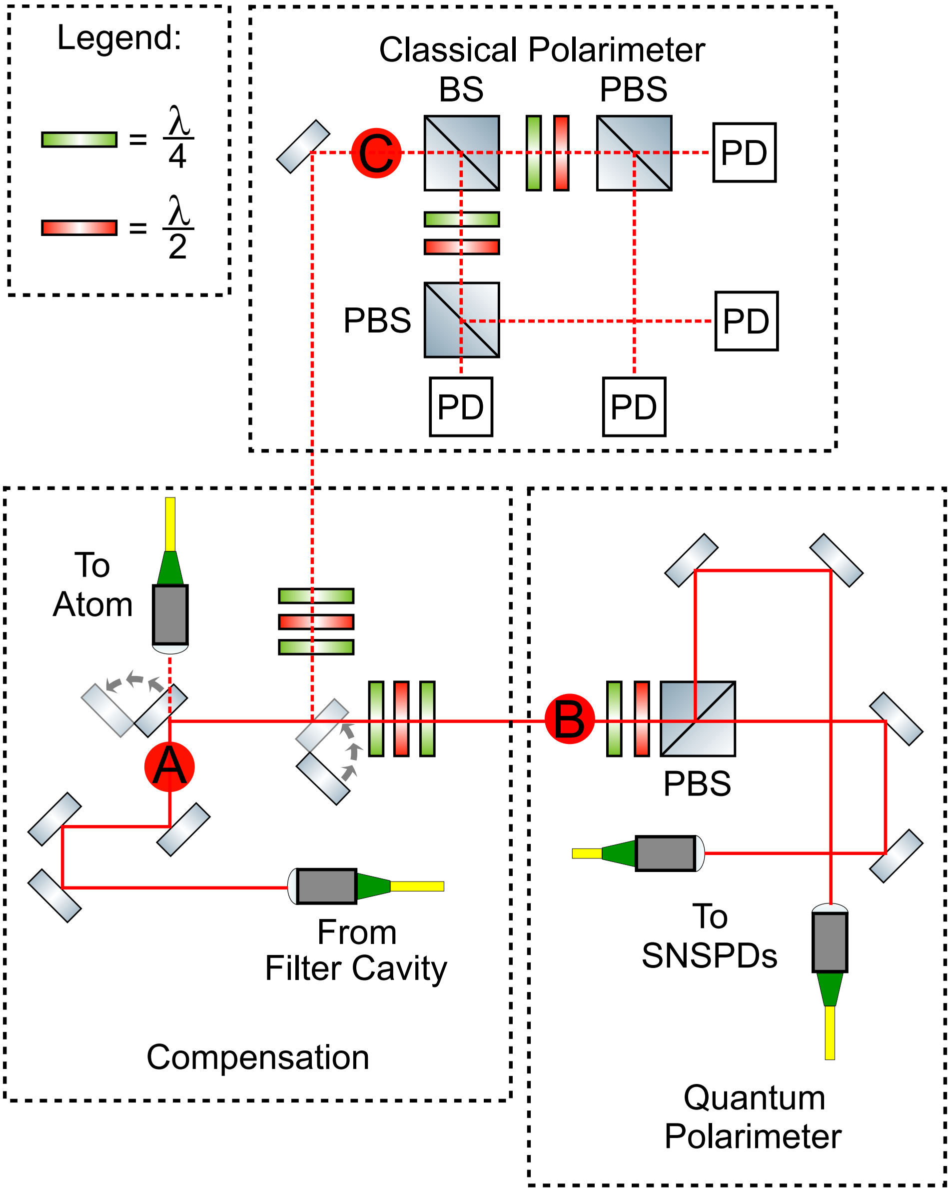



Figure 4.5: This figure shows overall setup of the detection setup. It can be divided into three parts: One is the **compensation**, where all beam paths are compensated to a common reference point shown as the red dot named A using a combination of waveplates. The other is the **classical polarimeter** that can be used by flipping a flip mirror to analyse the polarization of classical light. The last part is the **quantum polarimeter** that analyses a single photon's polarization by using a combination of waveplates, polarizing beam splitter and SNSPDs.

## 4.2 Theoretical Model

In this section a theoretical model of the atom-photon entanglement experiment is developed that can later be compared to the experiment to get an understanding of whether the experiment is running as expected.
This section is structured as follows: First, a model to calculate the entangling rate is presented in section 4.2.1. Afterwards, a model describing the influences of noise on the fidelity of the entangled atom-photon state, where the state was already introduced in section 2.3, is presented in section 4.2.2. More importantly, the filter cavity setup as well as the quantum frequency converter are discussed with regards to their influence on the entangling rate and fidelity.

### 4.2.1 Entangling Rate

The entangling rate $r$ at which entanglement is established between the different nodes, is given by

$$r = \eta \cdot D \cdot f. \tag{4.1}$$

Here $\eta$ is the probability or efficiency of an experimental attempt to succeed. For example, this includes the probability for the entangled photon to be collected into the fiber at the start of the experiment at LMU as well as the probability to detect the photon on the superconducting nanowire single-photon detectors at the end of the experiment at MPQ.
The quantity $D$ is the duty cycle, which is the fraction of time of the total experimental run time, atom-photon entanglement is tried. For example, if only half of the total time an atom is present in the trap, where the presence of a trapped atom is necessary in order to emit an entangled photon, and the other half of the time there is no atom present such that an atom-photon entanglement try can not be started, the duty cycle would be $D = 0.5$.
$f = \frac{1}{T}$ denotes the frequency at which atom-photon entanglement distribution between LMU and MPQ is attempted. This quantity is also called the repetition rate. Here $T$ is the total time it takes to establish atom-photon entanglement in a successful run. For example, the time it takes for the photon to travel from LMU to MPQ critically impacts the total time of the experiment, which thus limits the repetition rate. Another example is the time it takes to prepare the atom in a certain initial state.

The complete information regarding the entangling rate from equation 4.1, can be found in table 4.1, which displays efficiencies of the experimental devices, table 4.2 which displays the duration of different experimental processes as well as the resulting repetition rate and table 4.3, which shows the duty cycle of different experimental processes.
In the following, the structure of each table is quickly explained. The leftmost column of each table contains devices or processes that affect the overall efficiency, duration or duty cycle of the experiment. The next column indicates whether this device or process is at MPQ or LMU or both. The last column contains the relevant value for the efficiency, duration or duty cycle of that device or process. The last row is then the total efficiency, total duration or total duty cycle of the experiment, which is obtained by either the sum or multiplication of all rows

| Device or Process | Institution | Efficiency |
|---|---|---|
| Photon Collection | LMU | $0.01$ |
| MEMS | LMU | $0.85$ |
| Quantum Frequency Converter I | LMU | $0.35$ |
| Quantum Channel | LMU & MPQ | $0.11$ |
| Quantum Frequency Converter II | MPQ | $0.43$ |
| Filter Cavity Setup | MPQ | $0.70$ |
| Projection Setup | MPQ | $0.79$ |
| Fiber Connectors | MPQ | $0.92$ |
| SNSPDs | MPQ | $0.8$ |
| Post Processing | MPQ | $0.54$ |
| Whole Experiment | LMU & MPQ | $0.00003$ |

Table 4.1: Conservative estimates of specific efficiencies of singular devices appearing on the pathway of a photon between LMU and MPQ. The overall efficiency is given by multiplying all values in the last column except the last row to $\eta = 0.00003$, meaning that $1$ in roughly $33,300$ photons arrive at MPQ.

| Device or Process | Institution | Duration in $\mu$s |
|---|---|---|
| Qubit Preparation | LMU | $3$ |
| Entangling Pulse | LMU | $0.2$ |
| Raman Transfer | LMU | $8$ |
| Photon Travel Time | LMU & MPQ | $118.7$ |
| Signal Travel Time | LMU & MPQ | $118.7$ |
| Whole Experiment | LMU & MPQ | $248.6$ |

Table 4.2: Duration of processes in the experiment. The overall duration of one experiment is given by summing up all values in the last column except the last row yielding $T = 248.6\,\mu$s, which means that the repetition rate is $f = 4023\,$Hz.

| Device or Process | Institution | Duty Cycle |
|---|---|---|
| Atom Trapped | LMU | $0.8$ |
| Verify Trapping | LMU | $0.83$ |
| Atom Cooling & Magnetic Stabilization | LMU | $0.22$ |
| Polarization Compensation | LMU & MPQ | $0.95$ |
| Whole Experiment | LMU & MPQ | $0.14$ |

Table 4.3: Specific duty cycles of processes appearing in the experiment. The overall duty cycle of the experiment is given by multiplying all values in the last column except the last row, yielding $D = 0.14$. This means that during $14\,\%$ of the whole experimental run time, the generation of atom-photon entanglement is attempted.

above the last.
Using the value given for the efficiency $\eta$ in table 4.1, duty cycle $D$ in table 4.3 and repetition rate $f$ from table 4.2 and plugging all of them into equation 4.1, the theoretically predicted entangling rate $r_{\text{th}}$ is:

$$r_{\text{th}} = 1.04\,\text{min}^{-1}$$

This means that entanglement between an atom and a photon is established about one time a minute.

### 4.2.2 Entanglement Fidelity

In this section, a theoretical model is presented on how the initial experimental atom-photon state is influenced by the different devices and processes of the experiment.
The initial state at LMU is modelled as explained for equation 2.28

$$\rho_i = V\left|\phi^+\right\rangle\left\langle\phi^+\right| + (1-V)\frac{\mathbb{1}^{6\times 6}}{6},$$

where $\rho_i$ is the initial density matrix before any noise process is applied.
The photonic part of this state is then transmitted to MPQ, where it is measured. However, due to different noise contributions the photonic state is degraded. The probability that a noise photon is detected is called $p_{\text{n}}$, while the probability that a signal photon is detected is called $p_{\text{s}}$ for each experimental try. Just as in section 2.3, the noise on the photonic part of the state is assumed to be modelled by white noise. To do so, a completely depolarizing channel is applied to the photonic part of the composite state that maps the state towards white noise. This depolarizing channel is given by

$$\epsilon(\rho) = \frac{1}{4}\sum_{j=0}^{3}(\mathbb{1}^{3\times 3}\otimes\sigma_j)\rho(\mathbb{1}^{3\times 3}\otimes\sigma_j), \tag{4.2}$$

where $\sigma_i$ are the Pauli matrices and $\mathbb{1}^{3x3}$ is the identity matrix on the atomic subspace. The final state is then modelled as follows:

$$\begin{aligned}\rho_f &= \Big(p_{\text{s}}\rho_i + p_{\text{n}}\epsilon(\rho_i)\Big)\frac{1}{p_{\text{s}}+p_{\text{n}}}\\ \rho_f &= \Big(p_{\text{s}}\rho_i + p_{\text{n}}\frac{1}{4}\sum_{j=0}^{3}(\mathbb{1}^{3\times 3}\otimes\sigma_j)\rho_i(\mathbb{1}^{3\times 3}\otimes\sigma_j)\Big)\frac{1}{p_{\text{s}}+p_{\text{n}}},\end{aligned} \tag{4.3}$$

where $\rho_f$ is the final state and $\frac{1}{p_{\text{s}}+p_{\text{n}}}$ is used for normalizing the density matrix since $p_{\text{s}}+p_{\text{n}} \neq 1$ as there are experiments, where neither a noise nor a signal photon is detected.
For the following discussion of the fidelity, the duty cycle will be neglected, since the experiments can be postselected on whether atom-photon entanglement is being tried or not to

mitigate noise. This is possible, as it is known for example when the polarization compensation is running instead of an atom-photon entanglement try. The corresponding postselected signal to noise ratio ($\mathrm{SNR}$) can be defined as follows:

$$\mathrm{SNR} = \frac{p_{\mathrm{s}}}{p_{\mathrm{n}}}, \tag{4.4}$$

where $p_{\mathrm{s}} = \eta$ since it is the probability that a photon emitted from LMU is detected at MPQ. Using this definition of the SNR, the final state from equation 4.3 can be simplified to

$$\rho_f = \frac{\mathrm{SNR}}{\mathrm{SNR}+1}\rho_i + \frac{1}{\mathrm{SNR}+1}\frac{1}{4}\sum_{j=0}^{3}(\mathbb{1}^{3\times 3}\otimes\sigma_j)\rho_i(\mathbb{1}^{3\times 3}\otimes\sigma_j), \tag{4.5}$$

which can be verified by plugging in equation 4.4 to find equation 4.3.
Therefore, the only experimental variable that is left, which describes the final shared entangled state between LMU and MPQ, is the signal to noise ratio. The fidelity according to equation 3.16 between the final state $\rho_f$ from equation 4.3 and the ideal state $|\phi^+\rangle$ from equation 2.14 that should be prepared, is given by

$$F = \left\langle \phi^+ \right| \rho_f \left| \phi^+ \right\rangle. \tag{4.6}$$

This equation is not simplified further, since the QuTiP library in Python is used to illustrate the fidelity as a function of the signal to noise ratio. The fidelity from equation 4.6 is plotted versus the $\mathrm{SNR}$ in figure 4.6 by the blue curve labelled theoretical model. The other curve in red labelled theoretical limit, is the value that is approached by the model.
The curve of the theoretical model starts at $F = 0.235$ for $\mathrm{SNR} = 0$. Then it increases quickly until about $\mathrm{SNR} \approx 5$, where it starts to saturate. This shape can be explained as follows: One of the terms contributing to the fidelity is $\frac{\mathrm{SNR}}{\mathrm{SNR}+1}$, which has a similar shape to the curve in this figure[2].
Moreover, the value that is approached by the model is $F = 0.85$. This value is given by the fact that the best possible visibility for atom-photon correlations is $V = 0.82$, which is measured in-house at LMU. Therefore as the signal to noise approaches infinity, the fidelity from equation 4.6 simplifies to $F_{lim} = \frac{5}{6}V + \frac{1}{6}$ just like calculated in equation 2.28. Plugging in $V = 0.82$, yields the aforementioned limit.

To estimate the $\mathrm{SNR}$, a model that takes into account the experimentally relevant parameters is developed in the following. To reiterate, the $\mathrm{SNR}$ is given by equation 4.4

$$\mathrm{SNR} = \frac{p_{\mathrm{s}}}{p_{\mathrm{n}}},$$

[2]This becomes immediately obvious by explicitly writing down $\rho_f$ in equation 4.6

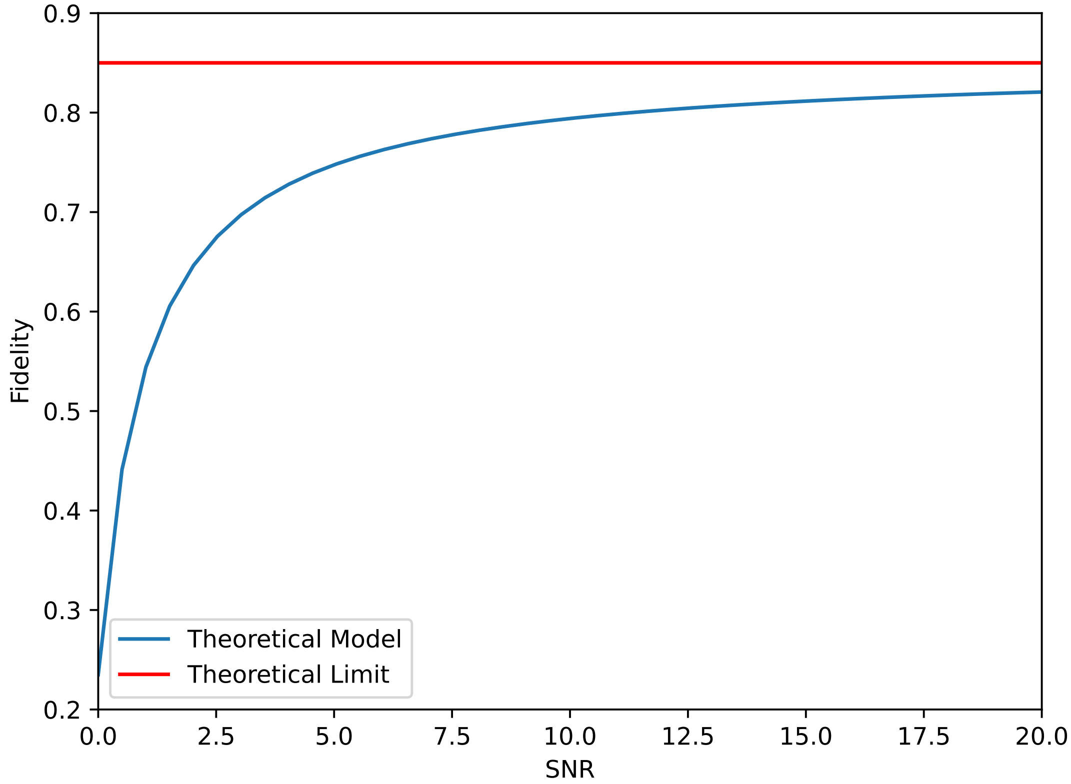



Figure 4.6: The fidelity on the vertical axis is plotted versus the $\mathrm{SNR}$ on the horizontal axis. The curve in blue is the theoretical model of the fidelity according to equation 4.6, while the red line is the theoretical limit when the $\mathrm{SNR}$ approaches infinity. The curve starts to saturate around $\mathrm{SNR} \approx 5$. Therefore any value above $5$ shows a good fidelity and improvements in the $\mathrm{SNR}$ will only result in small changes in fidelity.

with $p_\mathrm{s} = \eta$, where $\eta$ is the overall experimental efficiency as given in table 4.1. The probability to detect noise in the experiment $p_\mathrm{n}$ is modelled as follows:

$$p_\mathrm{n} = \left(\dot{n}_\mathrm{Lock} + \dot{n}_\mathrm{SNSPD} + \dot{N}_\mathrm{QFC} \cdot \mathrm{FWHM}\right)\Delta t, \tag{4.7}$$

where $\dot{n}_\mathrm{Lock}$ is the noise rate on the detectors due to stray lock light as discussed in section 4.1.1. $\dot{n}_\mathrm{SNSPD}$ is the combined dark count rate of the SNSPDs. Moreover, $\dot{N}_\mathrm{QFC}$ is the noise rate from the QFC in $\frac{\mathrm{Hz}}{\mathrm{MHz}}$ and $\mathrm{FWHM}$ is the linewidth of the filter cavity. The time frame $\Delta t$ is the post-processing window in which photon events are accepted after LMU has successfully excited an atom. This window is the reason why only $54\,\%$ of events at MPQ are accepted after post-processing, see table 4.1. This window is chosen as a trade off between good $\mathrm{SNR}$ and statistically significant detection events. By making this window

smaller, the $\mathrm{SNR}$ and thus fidelity can be increased.
In the experiment both $\eta$ and $p_\mathrm{n}$ are mainly controlled by two experimental devices[3], which are:

1. Quantum frequency converter II
2. Filter Cavity Setup

This is the case, as both contribute to the overall efficiency $\eta$, see table 4.1, as well as the overall noise. The filter cavity contributes to the noise, because depending on its $\mathrm{FWHM}$ more or less noise from the quantum frequency converter is transmitted, see equation 4.7. The second quantum frequency converter contributes to the noise as depending on the optical pump power, it produces more or less noise per frequency.
The following noise is measured in the experiment:

1. $\dot{n}_\mathrm{SNSPD} = 0.55$ Hz
2. $\dot{n}_\mathrm{Lock} = 1.11$ Hz

while the post-processing window is set to $\Delta t = 60$ ns as a trade-off between $\mathrm{SNR}$ and data collection.

In the following, the second quantum frequency converter and the filter cavity setup are discussed in more detail with regards to their impact on entangling rate and signal to noise ratio, which ultimately determines the fidelity according to equation 4.5. One needs to find a balanced trade-off in the settings of each device, where the entangling rate is high enough such that the timescales to gather enough data are deemed reasonable and where the fidelity, so the $\mathrm{SNR}$, is high enough such that the state can be proven to be entangled.

Figure 4.7 shows the $\mathrm{SNR}$ in blue and the entangling rate in green as a function of the filter cavity's $\mathrm{FWHM}$. The vertical line shows the working point in the experiment at $\mathrm{FWHM} = 32.8$ MHz. The entangling rate is calculated using equation 4.1, where the total efficiency $\eta$ is dependent on the cavity's $\mathrm{FWHM}$, since it changes the photon's transmission probability. Additionally, the $\mathrm{SNR}$ is calculated using the noise model from equation 4.7, where the quantum frequency converter is set such that the noise rate per frequency becomes $\dot{N}_\mathrm{QFC} = 1.12\,\frac{\mathrm{Hz}}{\mathrm{MHz}}$. This number is discussed in figure 4.8, which is found later in this section.
The entangling rate starts at $0$ for $\mathrm{FWHM} = 0$, since the cavity has no transmission at this point. However, the entangling rate cannot be observed to go to zero in figure 4.7 since the $\mathrm{FWHM}$ is plotted logarithmically. Moreover, the entangling rate can be seen to saturate at around $\mathrm{FWHM} \approx 100$ MHz. This happens as the photon's spectrum is smaller than $\approx 100$ MHz and therefore if the $\mathrm{FWHM}$ is increased beyond the photon's spectral width, the transmission does not change anymore.
The $\mathrm{SNR}$ can be seen to first increase with the $\mathrm{FWHM}$ of the cavity, as the overlap of the cavity's and the photon's spectral shape increases. Then, at around $\mathrm{FWHM} \approx 3$ MHz, the $\mathrm{SNR}$ has a maximum and accordingly starts to decrease afterwards. This comes from the

[3]Here the noise from the first quantum frequency converter is neglected, since it is highly attenuated through the fiber

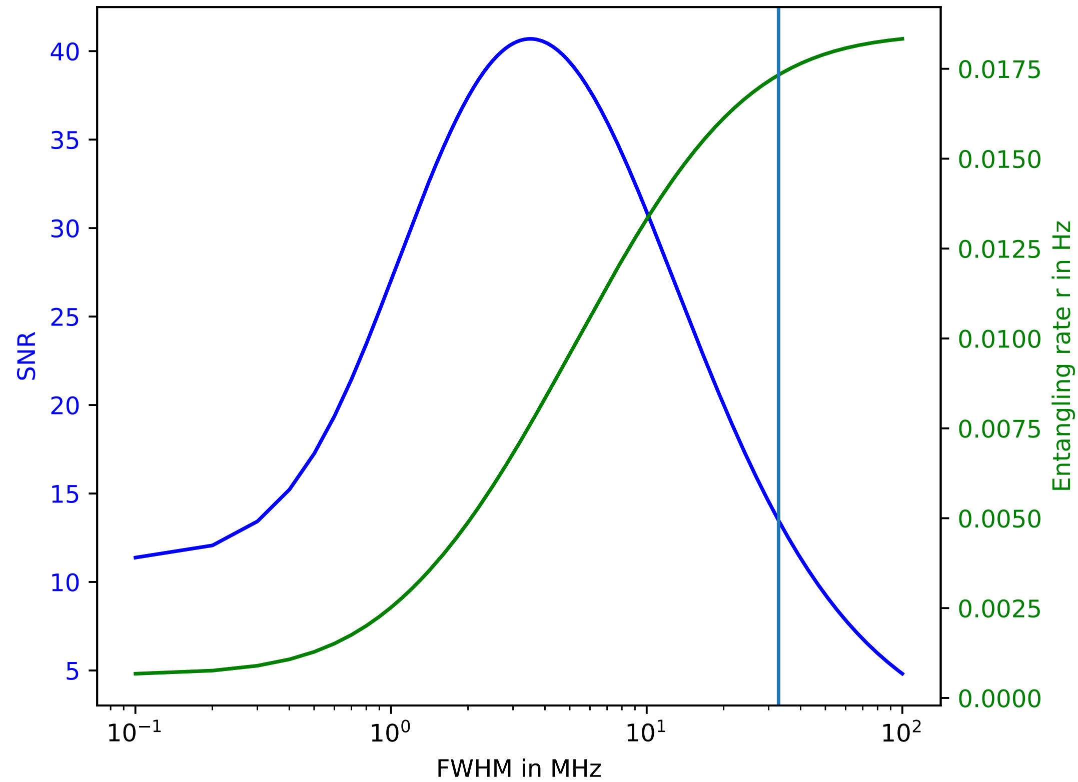



Figure 4.7: The $\mathrm{SNR}$ in blue and the entangling rate in green are plotted as a function of the filter cavity's linewidth in MHz. The vertical line shows the working point of $\mathrm{FWHM} = 32.8\,$MHz in the experiment. Using this figure, a trade-off of $\mathrm{SNR} = 13.47$ corresponding to $F = 0.807$ according to equation 4.6 and $r = 1.04\,\mathrm{min}^{-1}$ is predicted.

fact that the noise coming from the QFC increases faster than the transmitted photons with higher linewidths of the filter cavity. To find a trade-off between good $\mathrm{SNR}$ and entangling rate $r$, the cavity works at $\mathrm{FWHM} = 32.8\,$MHz.
Therefore, an $\mathrm{SNR} = 13.47$ corresponding to $F_{\mathrm{th}} = 0.807$ according to equation 4.6 and entangling rate of $r_{\mathrm{th}} = 1.04\,\mathrm{min}^{-1}$ are predicted from figure 4.7.

Figure 4.8 shows the $\mathrm{SNR}$ in blue and entangling rate in green plotted versus the optical pump power in W that is incident on the nonlinear crystal in the second quantum frequency converter. The entangling rate is again calculated using equation 4.1, where it is down to depend on the conversion efficiency, which is measured as a function of the optical pump power. The $\mathrm{SNR}$, which is also dependent on the conversion efficiency, is calculated using equation 4.7, where the cavity's linewidth is fixed to the working point of $\mathrm{FWHM} = 32.8\,$MHz as discussed for figure 4.7. Additionally, the noise from the quantum frequency converter, which also contributes to the $\mathrm{SNR}$, is measured in dependence of the optical pump power.

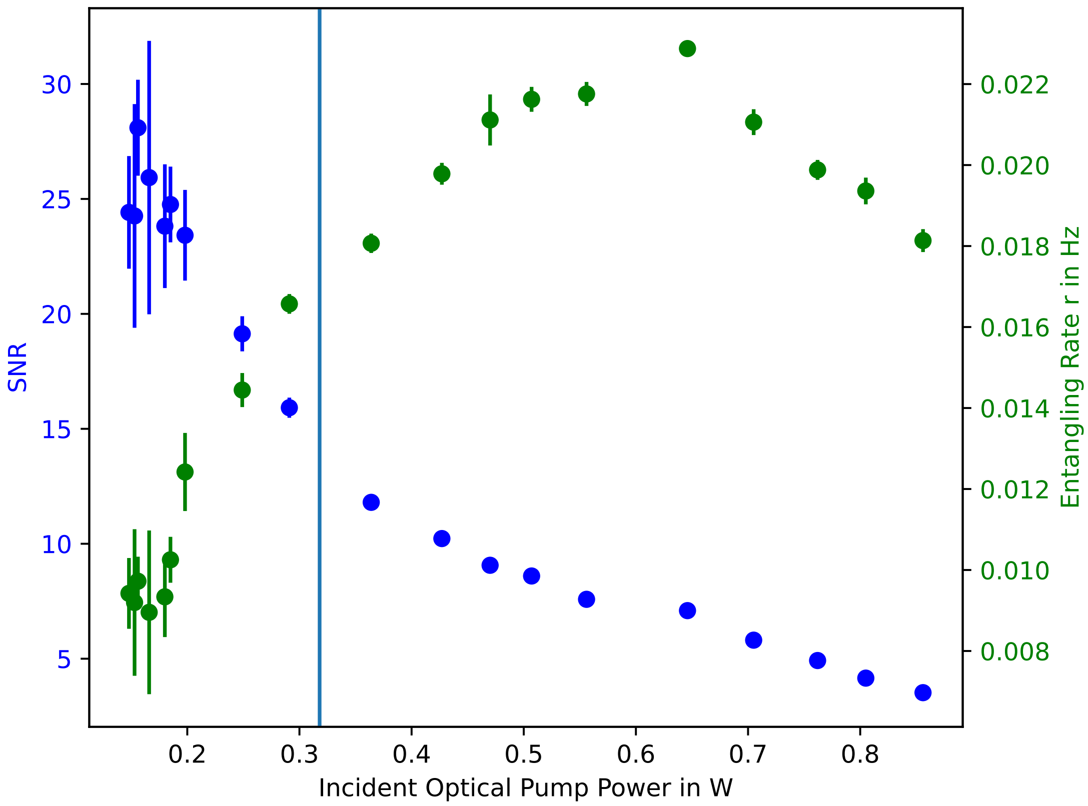



Figure 4.8: The $\mathrm{SNR}$ in blue and the entangling rate in green are plotted as a function of the optical pump power in W incident on the crystal in the quantum frequency converter. The vertical line shows the experimental working point of $318$ mW, which corresponds to a current of $950$ mA supplied to the amplifier. Using this figure, a trade-off of $\mathrm{SNR} = 13.47$ corresponding to $F = 0.807$ according to equation 4.6 and $r = 1.04\,\mathrm{min}^{-1}$ is predicted.

The optical pump power itself is changed by sending it through an amplifier, whose current is varied to amplify the optical power. In the range of $100$ mA and $700$ mA, the amplifier's amplification does not change, which is why there is a larger amount of data points for the optical pump power at around $150$ mW.

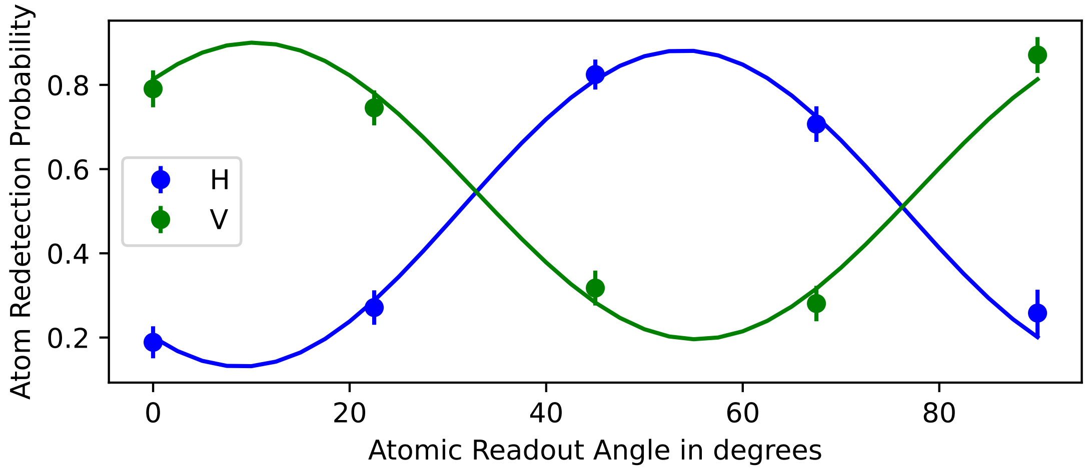
The error for the $\mathrm{SNR}$ and entangling rate are inferred by Gaussian error propagation from the measurements of the incident and converted power. For the $\mathrm{SNR}$ the error in the measurement of the noise also contributes to Gaussian error propagation. Systematic errors, such as errors in the incident optical power to the nonlinear crystal due to thermalization drifts of the amplifier, are not accounted for. The errors for the $\mathrm{SNR}$ and entangling rate are quite high for low optical pump power, since the laser that was being converted to measure the conversion efficiency, drifted quite fast in the beginning. The vertical line in light blue shows the optical pump power used in the experiment of $318$ mW, which corresponds to an

amplifier current of $950$ mA.
The entangling rate starts at about $0.009$ Hz for an optical pump power of about $150\,\mu$W. Then by increasing the optical pump power, the entangling rate starts to increase as the conversion efficiency increases. The increase in entangling rate stops at an optical pump power of about $650\mu$ W, where it starts to drop, since the conversion becomes less efficient. The $\mathrm{SNR}$ starts at about $23$ for an optical pump power of about $150\,\mu$W. With increasing optical pump power, the $\mathrm{SNR}$ can be seen to monotonously decrease. This is explained by the fact that the additionally converted photons increase slower than the additional noise photons with increasing optical pump power.
Once again a trade-off between good entangling rate and fidelity of the entangled state is found, which are the same values as discussed for 4.7.

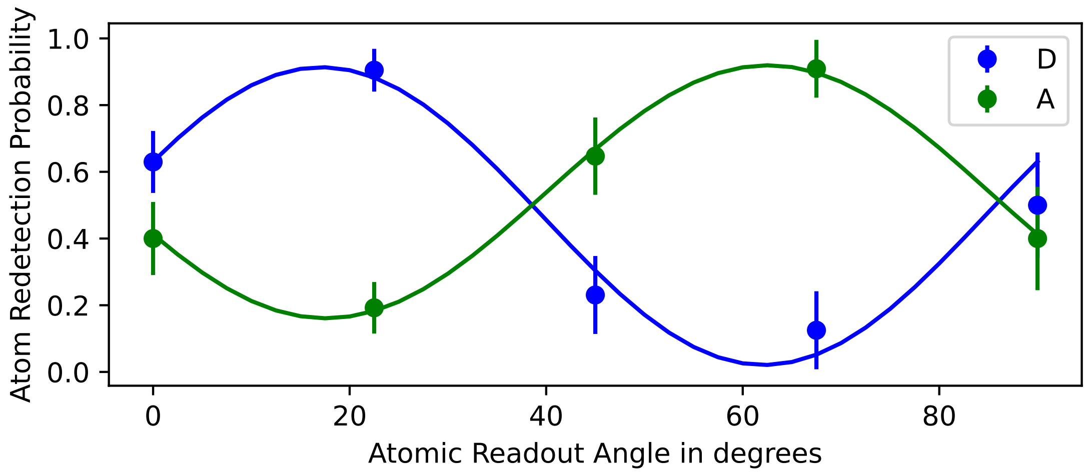
## 4.3 Results and Discussion

After having discussed the different parts of the setup as well as theoretical expectations in the section before, the fidelity of atom-photon entanglement will be analysed in the following using correlation measurements in two bases, also colloquially referred to as "fringe measurements". These are measurements of local operators, where one experimental parameter is varied, just as discussed in section 2.2. For the measurements presented below, the atomic analysis angle in $^\circ$ is varied. The visibility will be extracted from these measurements and according to equation 2.28, the fidelity of the atom-photon state is calculated. If it exceeds $50\,\%$ in both bases, the state is entangled according to section 2.4.

In figure 4.9, there are two plots. Figure 4.9a shows the atom-photon correlations by varying the atomic readout angle in $^\circ$ while the photon is measured in the H/V-basis. Figure 4.9b shows the same for the photonic D/A-basis. The vertical axis displays the atomic redetection probability, since LMU uses a destructive ionization readout, where the atom is ionized depending on its state using polarized light. In this scheme, the atom is either kicked out of the trap when ionized or stays trapped when not ionized, allowing for a projective measurement of the atomic state when detecting the ionized atom and electron on the walls of the surrounding vacuum chamber. The reader is referred to section 2.4.2. in [39] for more details of the atomic readout.
The atom is only read out once the photon has travelled the full path to MPQ, is detected using the quantum polarimeter and the resulting signal of the detection event is registered back again at LMU. This is done this way such that real entanglement is distributed over the quantum link instead of just classically correlated states by reading out the atom instantly. It needs to be mentioned that the active polarization stabilization did not work for both measurements. However, due to the high passive stability of the fiber, entanglement between the atom and photon could still be verified as can be seen later in the analysis of this chapter.
The points of each curve are generated as follows: For each entanglement try, the photon is either successfully transmitted and detected at MPQ or lost along the way. If it is successfully detected, an electronic signal heralding the detection event is sent back to LMU over the classical channel. When LMU receives this signal within the acceptance window $\Delta t$, the




(a) H/V-basis




(b) D/A-basis

Figure 4.9: Atom-Photon correlations for varying atomic readout angle for two different photonic bases. Figure 4.9a shows the atom-photon correlations in H/V- and figure 4.9b in D/A-basis. By fitting these with a sinusoidal function 4.8, the average visibility of these functions is found to be $\bar{V} = 0.777 \pm 0.038$. Using section 2.4, the fidelity is estimated to be $F = 0.814 \pm 0.031$.

atomic state at LMU is also readout and this contributes to a point in figure 4.9. Depending on whether the photon's polarization was measured in the H/V- or D/A-basis, the point either contributes to figure 4.9a or figure 4.9b respectively. The error of each point is given by the uncertainty due to the limited number of detection events.
These plots are fit with a sinusoidal function as the correlations are expected to oscillate

sinusoidally with the atomic readout angle, as shown for example in section 2.4. Therefore, the following function is used for fitting:

$$V \cdot \sin\Big(2\theta\frac{\pi}{180°} + \phi\Big)^2 + n, \tag{4.8}$$

where V is the visibility, which is the parameter of interest to estimate a lower bound for the fidelity according to equation 2.28. The variable $\theta$ is the atomic analysis angle in $°$ - converted to radiant using $\frac{\pi}{180°}$. The parameter $\phi$ is the offset of the atomic analysis angle and $n$ is the vertical offset for the correlations.The phase offset $\phi$ occurs in the experiment due to different experimental reasons such as incorrect basis calibration or incorrect polarization compensation.

The angular frequency of the oscillation is not a parameter of the fit. This is because polarized light is used to readout the atom, which is controlled by a $\lambda/2$-waveplate. If the polarization is aligned with the fast axis at $0°$ of the waveplate and the waveplate is rotated from $0°$ to $90°$, light used for readout of the atom is shifted by $180°$ in polarization. This corresponds to a readout of the atom in the anti-parallel basis and therefore the sign of the correlation flips. Thus the angular frequency needs to be $2$, since half of an oscillation of the correlations is completed in $90°$ instead of $180°$ as argued above.

By fitting both curves from figure 4.9a and figure 4.9b with equation 4.8 respectively and taking the errors in the redetection probability into account, the following visibilities are extracted:

$$V_H = 0.751 \pm 0.002 \tag{4.9}$$
$$V_V = 0.705 \pm 0.006 \tag{4.10}$$
$$V_D = 0.893 \pm 0.014 \tag{4.11}$$
$$V_A = 0.759 \pm 0.001 \tag{4.12}$$

The visibility $V_D$ of the correlation measurement when a diagonally polarized photon is detected, differs quite a lot from the other visibilities. Its value is most likely higher due to the lower amount of data points, which makes the fit less certain.

By averaging over the visibilities obtained from the fits, the following mean visibility is obtained

$$\bar{V} = 0.777 \pm 0.038,$$

where the error is obtained using Gaussian error propagation. Using this value and plugging it into the lower bound for the fidelity 2.28, the following mean fidelity is obtained

$$\bar{F} = 0.814 \pm 0.031.$$

This fidelity clearly exceeds the criterion of $F > 0.5$ in two different bases to witness entanglement, even when including the error.

Additionally, the atom-photon entanglement is detected at a rate of $r = (1.014 \pm 0.053)\,\mathrm{min}^{-1}$ for figure 4.9b, which is about one time per minute. A similar rate is also measured for figure

4.9a. The error in the entangling rate is given by Gaussian error propagation because of the statistical uncertainty in the amount of detected photons.

Comparing the fidelity as well as entangling rate measured in the experiment with the theoretically calculated values from section 4.2, $F_{\text{th}} = 0.807$ and $r_{\text{th}} = 1.04\,\text{min}^{-1}$, shows very good within the error bars.

## 4.4 Summary and Literature Review

At the start of this chapter, the setup to distribute atom-photon entanglement in the quantum link was explained. Specifically, the filter cavity setup built within the scope of this thesis was characterized in more detail. It was measured to have an internal efficiency of $T_{in}(91.57 \pm 0.94)\,\%$. More importantly, an external efficiency of $T_{ext} = (75.14 \pm 1.17)\,\%$ was obtained which is mainly limited by low fiber coupling of $T_{fc} = (82.06 \pm 0.96)\,\%$ out of the device. One can expect to be able increase the external efficiency as high as $T_{ext} = 82\,\%$ if one were to use a telescope to increase the fiber coupling to $90\,\%$. Additionally, a cavity transmission of $T_{cav} = (97.7 \pm 1)\,\%$ was inferred, which is a good value compared to a cavity with similar design specifications that has a transmission of $T = 96.5\,\%$ [41].

In the atom-photon entanglement distribution experiment, the fidelity and entangling rate were measured to be $\bar{F} = 0.814 \pm 0.031$ and $r = (1.014 \pm 0.053)\,\text{min}^{-1}$ respectively. Moreover, these were found to be in good agreement with the theoretical predictions of $F_{\text{th}} = 0.807$ and $r_{\text{th}} = 1.04\,\text{min}^{-1}$, which were predicted using all the parameters laid out in section 4.2.
Since the fidelity in two different bases exceeded the criterion of $0.5$ to witness entanglement, the experimental atom photon state was verified to be entangled. .

| Experiment | Platform | $L$ in km | $r$ in Hz | F in % |
|---|---|---|---|---|
| LMU & MPQ | $^{87}\text{Rb}$ atom-photon | $23.7$ | $0.017(1)$ | $(81.4 \pm 3.1)$ |
| LMU [24] | $^{87}\text{Rb}$ atom-photon | $101$ | $0.004$ | $(70.8 \pm 2.4)$ |
| IQOQI [45] | multimode $^{40}\text{Ca}^{+}$ ion-photon | $101$ | $2.85(7)$ | $(87.7 \pm 0.5)$ |
| IQOQI [45] | $^{40}\text{Ca}^{+}$ ion-photon | $101$ | $1.23(5)$ | $(88 \pm 3)$ |
| Saarland [33] | $^{40}\text{Ca}^{+}$ ion-photon | $14$ | ? | $(79 \pm 2)$ |
| ICFO [46] | $\text{Pr}^{3+}$ in $\text{Y}_2\text{SiO}_5$-photon | $50$ | ? | $(88 \pm 3)$ |
| Harvard [47] | NV-centre in Diamond-photon | In-house | $\approx 1$ | $(69 \pm 6.8)$ |

Table 4.4: This table compares the travelled distance $L$ of the photon, the entangling rate $r$, and the stationary memory qubit-photon entanglement fidelity $F$ of different entanglement distribution experiments. A question mark is used, where values are unknown.

In table 4.4, different experiments are compared, where entanglement between some stationary memory qubit, e.g. a single atom or defect inside a solid-state, with a flying qubit is verified using the fidelity of the aforementioned state. There are many more experiments

that already establish entanglement between pairs of stationary memory qubits such as the work between The Hague and Delft in Netherlands [36] or even a three stationary memory qubit quantum network in Hefei in China [38]. However, these are not included here as they will be used as a comparison once atom-atom entanglement has been established.
The experiments shown in table 4.4 are performed at IQOQI in Austria [45], at Saarland University in Germany [33], at ICFO in Spain [46] and at Harvard University in the United States [47].
When comparing the length, the quantum link presented in this thesis of roughly $24\,\text{km}$ length does not immediately stand out in comparison to the other links, since some operate at distances of $50\,\text{km}$ or even $101\,\text{km}$. However, in the experiments from [24] and [45] spooled fibers in the laboratory are used. Therefore, the atom-photon entanglement distribution experiment shown in this thesis is technically more challenging, as it is performed at large distance with field deployed fibers. The entangling rate for the experiment presented here is really low compared to [45]. This is due to a technicality rather than a fundamental limit, as the low photon collection of $1\,\%$ at LMU seen in table 4.1 limits the overall rate significantly. By using a cavity, these values can be expected increase by almost a factor of $50$, which would raise the entangling rate to about $0.8\,\text{Hz}$. This rate would then be comparable to the rate of $1.23(5)$ in single ion experiment from [45].
The fidelity measured in the experiment was about $F = 81\,\%$, which compares well to the other experiments apart from [45], where they reach up to $88\,\%$ fidelity. The fidelity in this thesis is limited to $85\,\%$, because of the atomic decoherence during the experiment. This upper bound could be measured by eradicating different experimental constraints such as imperfect polarization compensation of the fiber or imperfect atomic and photonic readout, which are limiting the measured fidelity currently. If the upper bound fidelity of $85\,\%$ would be reached, the fidelity would would then compare well to [45].

## Chapter 5

# Conclusion and Outlook

To achieve the goal of building a quantum link between a $^{87}\mathrm{Rb}$-atom at LMU in the center of Munich and another $^{87}\mathrm{Rb}$-atom at MPQ in Garching, initial steps towards establishing atom-atom entanglement between the two nodes were described in this thesis. These include the characterization of the two optical telecom fibers as possible quantum channels and the distribution of an entangled photon emitted by the atom at LMU to MPQ.

## Optical Fiber Network

First, two optical telecom fibers were investigated using an optical time-domain reflectometer measurement. Using this measurement, the fibers were checked for faulty splices or losses, but no obvious signs of faults or damage were identified. Additionally, the the green fiber was determined to be $23.71$ km long while the red fiber was was determined to be $23.74$ km long. Using two calibrated power meters, the green fiber was measured to have an attenuation of $8.41$ dB while the red fiber was measured to have an attenuation of $9.14$ dB, which aligned with expectations. Therefore, the green fiber was chosen as the quantum channel to minimize losses of quantum information. The red fiber was then used for transmitting all kinds of classical signals such as TTL pulses, RF signals or Ethernet communication.
More importantly, the fibers were characterized regarding their polarization dependent losses as well as polarization drifts in time. These are important measures for quantum channels supporting qubits encoded in the polarization degree of photons, since they give insight into how the quantum information is preserved through the channel. The polarization dependent loss of the quantum channel was measured using the Müller-matrix method described in [32]. Essentially, one measures the transmission of four different input polarizations to infer the polarization dependent loss. The polarization dependent loss was found to be $(0.109 \pm 0.019)$ dB for the quantum and $(0.083 \pm 0.024)$ dB for the classical channel. The comparison to a polarization dependent loss of $(0.08 \pm 0.09)$ dB for a $40\,\%$ shorter fiber in Saarland [33], indicates that the polarization dependent losses for the fibers of this thesis are relatively low.

The polarization stability was measured by sending classical light through each fiber respectively for $48$ h and monitoring the polarization after the fiber. The stability was determined to be $\approx 25$ min for the classical channel and $\approx 100$ min for the quantum channel. The difference in stability was explained by influences from the experimenter such as accidentally touching the fiber.
Comparing to [33], [34], [36], [37] and [38], the fibers of this thesis showed the highest passive stability together with the fiber network in China [38]. Also comparing the length of the fibers from this thesis to other field deployed fibers, only the fibers in Boston [34] and are longer with a length of $42.5$ km. It was concluded that the fibers of this thesis show very good characteristics as quantum channels for qubits encoded in the polarization of single photons. The full comparison of different fibers can be found in table 3.2.

## Atom-Photon Entanglement

After the characterization of the optical fiber network, atom-photon entanglement distribution was investigated. To be able to do so, a filter cavity setup was built first to filter the noise coming from the second quantum frequency converter that converts photons from telecom to near infrared wavelengths. The external device efficiency of the filter cavity setup was determined to be $T_{ext} = (75.14 \pm 1.17)\,\%$. This number was found to be mainly limited by the fiber coupling efficiency of $T_{fc} = (82.06 \pm 0.96)\,\%$, possibly due to bad mode matching. The corresponding internal device efficiency was measured to be $T_{in} = (91.57 \pm 0.94)\,\%$, which was mainly limited by the reflection from the Volume-Bragg grating of $T_{VBG} = (96.39 \pm 0.2)\,\%$ and cavity transmission of $T_{cav} = (97.7 \pm 1)\,\%$. These figures were in line with expectations. Moreover, a good coupling to the TEM00 cavity mode was demonstrated by a side mode suppression of $> 24$ dB to the next highest higher-order mode.
Secondly, the entanglement between a photon at MPQ that was emitted by an atom at LMU was quantified by measuring atom-photon correlations between LMU and MPQ. By varying the atomic readout basis at LMU and measuring the atom as well as detecting the photon's polarization at MPQ in the H/V- and D/A-basis, oscillating correlations were observed with an average visibility of $\bar{V} = 0.777 \pm 0.038$. This corresponds to a fidelity of $\bar{F} = 0.814 \pm 0.031$, proving that the distributed state was entangled according to the entanglement criterion of $F > 0.5$, which needs to be verified in two different bases. Additionally, the entangling rate of the entanglement distribution was measured to be $r = (1.014 \pm 0.053)\,\mathrm{min}^{-1}$. These measurements were taken without active polarization stabilization, as it did not work yet. The fact that entanglement could be demonstrated shows how passively stable these fibers are. Moreover, the experimental values were found to agree with the theoretical predictions of $F_{\mathrm{th}} = 0.807$ and $r_{\mathrm{th}} = 1.04\,\mathrm{min}^{-1}$ when considering the error bars.
If one wants to increase the fidelity of the atom-photon state, while keeping the entangling rate roughly the same, reducing the cavity's length could be investigated, such that less additional noise spectra are transmitted in the future. Another method that was proposed to suppress the additional noise spectra, is to build a second cavity with a different finesse, where the $\mathrm{FWHM}$ stays the same but the $\mathrm{FSR}$ of the cavity is increased.

Moreover, other entanglement distribution experiments [24], [45], [33], [46] and [47] between a stationary memory qubit and a single photon were compared to this work in table 4.4. The highlight of the atom-photon entanglement distribution of this thesis is the use of a field deployed fiber between distant laboratories with good fidelity, which otherwise only [33] and [46] report. The entangling rate was found to be limited by the technical implementation of the photon collection at LMU in comparison to other experiments, which is not a fundamental limit. By coupling the atom to a cavity, the photon collection was predicted to increase by at least an order of magnitude.

## Outlook: Atom-Atom Entanglement

The next step for this project is to store the photon that is entangled with the atom at LMU into the atom at MPQ to establish atom-atom entanglement, thereby completing the quantum link's capabilities. A high-fidelity heralded storage and readout of a photonic qubit was already verified in the past for the MPQ node. The average state fidelity of the state after readout for storage times shorter than $25\,\mu$s was demonstrated to be $(94.7 \pm 0.2)\,\%$ [48]. Moreover, the storage efficiency, defined as the probability to detect the herald photon for an incident photonic qubit, was measured to be $(11 \pm 1)\,\%$ [48].

Additionally, using the resource of node to node entanglement in the link, different experiments to showcase potential applications of the quantum link or to investigate fundamental physics could be performed.
One such experiment could be to establish a secret key shared by LMU and MPQ using the E91 protocol [49]. For this protocol, many pairs of entangled atoms between LMU and MPQ have to be created and measured in random bases. Then, LMU and MPQ communicate classically to coordinate which share of atoms was measured using the same bases. This part of the entangled atoms is used to establish the secret key only when the other rest of entangled atoms can be used to violate Bell's inequalities as this implies that no eavesdropper had access to the shared quantum states.
Another practical application that could be investigated after atom-atom entanglement has been established is quantum teleportation [50], based on reflection of a photon of the cavity [51] at MPQ. By doing so, a Bell state measurement on the atom and reflected photon can be implemented. Communicating the outcome of said measurement to LMU, they can apply appropriate local operations to their atom, such that the state of the photon is teleported to LMU's atom.
An experiment in the direction of fundamental physics could be to investigate quantum steering between the nodes [52]. For steering, one party creates a bipartite quantum state, for example LMU, and shares it with another party, here MPQ. The task of LMU is now to convince MPQ, who believe quantum mechanics is correct but do not trust LMU, that the state they prepared is entangled. Therefore LMU succeeds, if the correlations of the composite state cannot be explained using a Local Hidden State model that models MPQ as having a definite quantum state even if they do not know it.

# Appendices

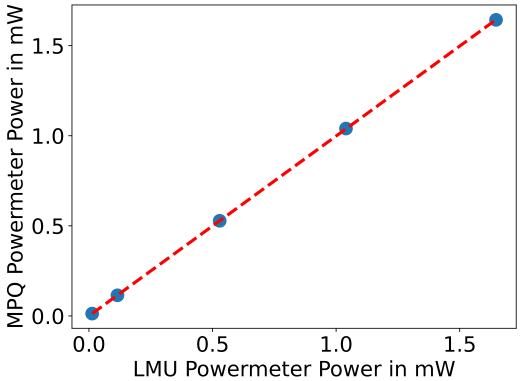
# Appendix A

# Power Meter Calibration

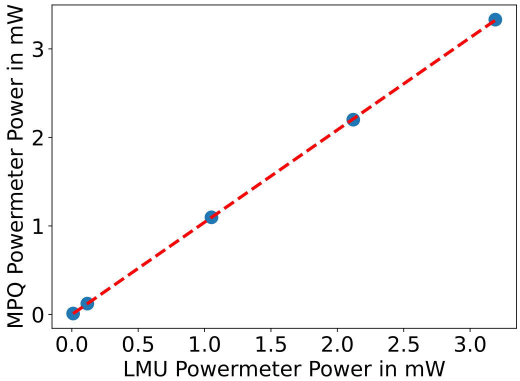
The calibration curves of the power meters can be found in figure A.1. Each plot shows the power measured by the MPQ power meter on the vertical axis and the power measured by the LMU power meter on the horizontal axis for three different wavelengths.

The power meter "PM100D" from Thorlabs was used in both cases. The MPQ power meter was used with a "S122" type power sensor, while the LMU power meter was used with a slim "S132-C" type power sensor. For the LMU power sensor, the more precise measuring mode that can go up to $5$ mW is used. The other mode would be able to measure powers up to 500mW with less precision.

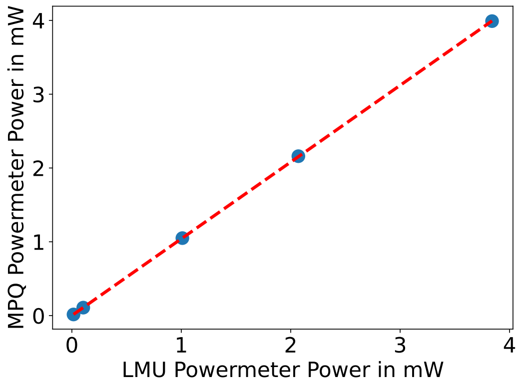
Both power sensors had a "SMF1CA" connector to convert from fiber to free space. Therefore, there is no uncertainty in either power, since Thorlabs only specifies uncertainties for beam diameters >1 mm at the power sensors. The beam diameter is always below this threshold as the light is guided within a "SMF-28 Ultra" single-mode fiber by Thorlabs, which has a specified mode diameter of approximately 10 µm. When this Gaussian-like mode is coupled into free space, its diameter is broadened over its propagation distance. However, the free space mode only reaches the threshold after approximately a centimeter of propagation length, which is much longer than the distance between fiber and power sensor that is set by the connector.




(a) This plot shows the power meter calibration for light with a wavelength of $1514$ nm. The slope of the linear fit is given by $m = 0.998$ and the vertical intercept by $n = -0.018$μW.




(b) This plot shows the power meter calibration for light with a wavelength of $1550$ nm. The slope of the linear fit is given by $m = 1.042$ and the vertical intercept by $n = -0.981$μW.




(c) This plot shows the power meter calibration for light with a wavelength of $1610$ nm. The slope of the linear fit is given by $m = 1.04$ and the vertical intercept by $n = 0.949$μW.

Figure A.1: This figure shows plots used for calibrating the power meter from MPQ and LMU with respect to each other. The vertical axis shows the power measured with the power meter from MPQ. The horizontal axis shows the exact same source measured with the power meter from LMU. The data in blue is fit with a linear function in red, which fits precisely to the data points.

# Abbreviations

| | |
|---|---|
| **LMU** | Ludwig Maximilian University |
| **MPQ** | Max-Planck-Institute of Quantum Optics |
| **PPT** | Partial positive transpose |
| **OTDR** | Optical time-domain reflectometer |
| **PDL** | Polarization dependent loss |
| **SNSPD** | Superconducting nanowire single-photon detector |
| **QFC** | Quantum frequency converter |
| **VBG** | Volume-Bragg grating |
| **PD** | Photodiode |
| **PID** | Proportional–integral–derivative |
| **AC** | Alternating current |
| **DC** | Direct current |
| **FSR** | Free spectral range |
| **FWHM** | Full width at half maximum |
| **SNR** | Signal to noise ratio |

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# Bibliography


[1] Walther Gerlach and Otto Stern. “Der experimentelle Nachweis der Richtungsquantelung im Magnetfeld”. In: *Zeitschrift für Physik* 9 (1 Dec. 1922), pp. 349–352. ISSN: 14346001. DOI: 10.1007/BF01326983/METRICS. URL: https://link.springer.com/article/10.1007/BF01326983.

[2] A Einstein. “The Quantum Theory of Radiation”. In: *Physikalische Zeitschrift* 18 (), p. 121.

[3] Jonathan P Dowling and Gerard J Milburn. “Quantum technology: the second quantum revolution”. In: *Philosophical Transactions of the Royal Society of London. Series A: Mathematical, Physical and Engineering Sciences* 361.1809 (2003), pp. 1655–1674.

[4] *Quantum Manifesto*. Tech. rep. European Union, 2016. URL: https://qt.eu/media/pdf/93056_Quantum-Manifesto_WEB.pdf (visited on 10/29/2024).

[5] Nicolas Gisin and Rob Thew. “Quantum communication”. In: *Nature photonics* 1.3 (2007), pp. 165–171.

[6] Christian L Degen, Friedemann Reinhard, and Paola Cappellaro. “Quantum sensing”. In: *Reviews of modern physics* 89.3 (2017), p. 035002.

[7] Thaddeus D Ladd et al. “Quantum computers”. In: *nature* 464.7285 (2010), pp. 45–53.

[8] Andreas Reiserer and Gerhard Rempe. “Cavity-based quantum networks with single atoms and optical photons”. In: *Reviews of Modern Physics* 87.4 (2015), pp. 1379–1418.

[9] Vittorio Giovannetti, Seth Lloyd, and Lorenzo Maccone. “Quantum-enhanced measurements: beating the standard quantum limit”. In: *Science* 306.5700 (2004), pp. 1330–1336. DOI: 10.1126/science.1104149.

[10] Marcello Caleffi et al. “Distributed quantum computing: a survey”. In: *arXiv preprint arXiv:2212.10609* (2022).

[11] Gláucia Murta et al. “Quantum Conference Key Agreement: A Review”. In: *Advanced Quantum Technologies* 3.11 (2020), p. 2000025. DOI: https://doi.org/10.1002/qute.202000025. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/qute.202000025. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/qute.202000025.

[12] Stephanie Wehner, David Elkouss, and Ronald Hanson. “Quantum internet: A vision for the road ahead”. In: *Science* 362.6412 (2018), eaam9288. DOI: 10.1126/science.aam9288. eprint: https://www.science.org/doi/pdf/10.1126/science.aam9288. URL: https://www.science.org/doi/abs/10.1126/science.aam9288.

[13] Loıc Henriet et al. “Quantum computing with neutral atoms”. In: *Quantum* 4 (2020), p. 327.

[14] Jacob P Covey, Harald Weinfurter, and Hannes Bernien. “Quantum networks with neutral atom processing nodes”. In: *npj Quantum Information* 9.1 (2023), p. 90.

[15] R Hughes et al. *A Quantum Information Science and Technology Roadmap, version2. 0*. Tech. rep. Report LA-UR-04-1778, Advanced Research and Development Activity (ARDA), 2004.

[16] Zheshen Zhang and Quntao Zhuang. “Distributed quantum sensing”. In: *Quantum Science and Technology* 6.4 (July 2021), p. 043001. DOI: 10.1088/2058-9565/abd4c3. URL: https://dx.doi.org/10.1088/2058-9565/abd4c3.

[17] Albert Einstein, Boris Podolsky, and Nathan Rosen. “Can quantum-mechanical description of physical reality be considered complete?” In: *Physical review* 47.10 (1935), p. 777.

[18] John S Bell. “On the einstein podolsky rosen paradox”. In: *Physics Physique Fizika* 1.3 (1964), p. 195.

[19] Nobel Prize Outreach AB. *The Nobel Prize in Physics 2022*. https://www.nobelprize.org/prizes/physics/2022/summary/. Accessed: 2024-11-17. 2024.

[20] Reinhard F. Werner. “Quantum states with Einstein-Podolsky-Rosen correlations admitting a hidden-variable model”. In: *Phys. Rev. A* 40 (8 Oct. 1989), pp. 4277–4281. DOI: 10.1103/PhysRevA.40.4277. URL: https://link.aps.org/doi/10.1103/PhysRevA.40.4277.

[21] Sirui Lu Ignacio Ciray. *QST Theory: Quantum Information*. Lecture Script. Mar. 2022.

[22] Nikolai Kiesel. “Experiments on multiphoton entanglement”. PhD thesis. lmu, 2007.

[23] Lukas Knips et al. “Multipartite entanglement analysis from random correlations”. In: *npj Quantum Information* 6.1 (2020), p. 51.

[24] Yiru Zhou et al. “Long-Lived Quantum Memory Enabling Atom-Photon Entanglement over 101 km of Telecom Fiber”. In: *PRX Quantum* 5 (2 Apr. 2024), p. 020307. DOI: 10.1103/PRXQuantum.5.020307. URL: https://link.aps.org/doi/10.1103/PRXQuantum.5.020307.

[25] Richard Jozsa. “Fidelity for mixed quantum states”. In: *Journal of modern optics* 41.12 (1994), pp. 2315–2323.

[26] Asher Peres. “Separability Criterion for Density Matrices”. In: *Phys. Rev. Lett.* 77 (8 Aug. 1996), pp. 1413–1415. DOI: 10.1103/PhysRevLett.77.1413. URL: https://link.aps.org/doi/10.1103/PhysRevLett.77.1413.

[27] Michał Horodecki, Paweł Horodecki, and Ryszard Horodecki. "Separability of mixed states: necessary and sufficient conditions". In: *Physics Letters A* 223.1 (1996), pp. 1–8. ISSN: 0375-9601. DOI: https://doi.org/10.1016/S0375-9601(96)00706-2. URL: https://www.sciencedirect.com/science/article/pii/S0375960196007062.

[28] Tim van Leent. "Entanglement of distant atoms for quantum networks". May 2023. URL: http://nbn-resolving.de/urn:nbn:de:bvb:19-318364.

[29] Tim van Leent et al. "Long-Distance Distribution of Atom-Photon Entanglement at Telecom Wavelength". In: *Phys. Rev. Lett.* 124 (1 Jan. 2020), p. 010510. DOI: 10.1103/PhysRevLett.124.010510. URL: https://link.aps.org/doi/10.1103/PhysRevLett.124.010510.

[30] David P. DiVincenzo. "The Physical Implementation of Quantum Computation". In: *Fortschritte der Physik* 48.9-11 (2000), pp. 771–783. DOI: https://doi.org/10.1002/1521-3978(200009)48:9/11<771::AID-PROP771>3.0.CO;2-E. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/1521-3978%28200009%2948%3A9/11%3C771%3A%3AAID-PROP771%3E3.0.CO%3B2-E. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/1521-3978%28200009%2948%3A9/11%3C771%3A%3AAID-PROP771%3E3.0.CO%3B2-E.

[31] William K Wootters and Wojciech H Zurek. "A single quantum cannot be cloned". In: *Nature* 299.5886 (1982), pp. 802–803.

[32] *Polarization-Dependent Loss (PDL) Measurement Theory*. Tech. rep. Scottsdale, Arizona, USA: VIAVI Solutions, 2015. URL: https://www.viavisolutions.com/en-us/literature/polarization-dependant-loss-pdl-measurement-theory-white-papers-books-en.pdf (visited on 08/27/2024).

[33] Stephan Kucera et al. *Demonstration of quantum network protocols over a 14-km urban fiber link*. 2024. arXiv: 2404.04958 [quant-ph]. URL: https://arxiv.org/abs/2404.04958.

[34] Eric Bersin et al. "Development of a Boston-area 50-km fiber quantum network testbed". In: *Phys. Rev. Appl.* 21 (1 Jan. 2024), p. 014024. DOI: 10.1103/PhysRevApplied.21.014024. URL: https://link.aps.org/doi/10.1103/PhysRevApplied.21.014024.

[35] Omar Gamel and Daniel FV James. "Measures of quantum state purity and classical degree of polarization". In: *Physical Review A—Atomic, Molecular, and Optical Physics* 86.3 (2012), p. 033830.

[36] Arian J. Stolk et al. *Metropolitan-scale heralded entanglement of solid-state qubits*. 2024. arXiv: 2404.03723 [quant-ph]. URL: https://arxiv.org/abs/2404.03723.

[37] Zi-Heng Xiang et al. "Long-term transmission of entangled photons from a single quantum dot over deployed fiber". In: *Scientific reports* 9.1 (2019), p. 4111.

[38] Jian-Long Liu et al. "Creation of memory–memory entanglement in a metropolitan quantum network". In: *Nature* 629.8012 (2024), pp. 579–585.

[39] Robert Sebastian Garthoff. "Efficient single photon collection for single atom quantum nodes". PhD thesis. Ludwig-Maximilians-Universität München, 2021.

[40] Jianming Huang and Prem Kumar. "Observation of quantum frequency conversion". In: *Physical review letters* 68.14 (1992), p. 2153.

[41] Matthias Bock. "Polarization-preserving quantum frequency conversion for trapped-atom based quantum networks". PhD thesis. 2020.

[42] Marcelo Alejandro Luda et al. "Compact embedded device for lock-in measurements and experiment active control". In: *Review of Scientific Instruments* 90.2 (Feb. 2019), p. 023106. ISSN: 0034-6748. DOI: 10.1063/1.5080345. eprint: https://pubs.aip.org/aip/rsi/article-pdf/doi/10.1063/1.5080345/15646092/023106_1_online.pdf. URL: https://doi.org/10.1063/1.5080345.

[43] Eric D. Black. "An introduction to Pound–Drever–Hall laser frequency stabilization". In: *American Journal of Physics* 69.1 (Jan. 2001), pp. 79–87. ISSN: 0002-9505. DOI: 10.1119/1.1286663. eprint: https://pubs.aip.org/aapt/ajp/article-pdf/69/1/79/10115998/79_1_online.pdf. URL: https://doi.org/10.1119/1.1286663.

[44] Wanlin Li. "Narrow-Band Spectral Filtering System". Preprint from August. Bachelor's Thesis. Ludwig-Maximilians-Universität, 2023.

[45] V Krutyanskiy et al. "Multimode ion-photon entanglement over 101 kilometers". In: *PRX Quantum* 5.2 (2024), p. 020308.

[46] Jelena V Rakonjac et al. "Transmission of light–matter entanglement over a metropolitan network". In: *Optica Quantum* 1.2 (2023), pp. 94–102.

[47] Emre Togan et al. "Quantum entanglement between an optical photon and a solid-state spin qubit". In: *Nature* 466.7307 (2010), pp. 730–734.

[48] Manuel Brekenfeld et al. "A quantum network node with crossed optical fibre cavities". In: *Nature Physics* 16.6 (2020), pp. 647–651.

[49] Artur K Ekert. "Quantum cryptography and Bell's theorem". In: *Quantum Measurements in Optics* (1991), pp. 413–418.

[50] Charles H Bennett et al. "Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels". In: *Physical review letters* 70.13 (1993), p. 1895.

[51] Andreas Reiserer et al. "A quantum gate between a flying optical photon and a single trapped atom". In: *Nature* 508.7495 (2014), pp. 237–240.

[52] Howard M Wiseman, Steve James Jones, and Andrew C Doherty. "Steering, entanglement, nonlocality, and the Einstein-Podolsky-Rosen paradox". In: *Physical review letters* 98.14 (2007), p. 140402.

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